

Task Accuracy Cannot Validate Activation Compression: Run-to-Run Variance and a Five-Minute Gradient Probe*

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September 28, 2026

Abstract

Fine-tuning a 70B model on a single 128 GB device without recomputation requires compressing the activations stored for the backward pass. Prior activation compression work validates such a method through an in-distribution task-accuracy change (0.5 points for ActNN, 1 point for GACT). We show that this check does not see the failure that four-bit compression actually produces. Compressing every stored tensor to four bits, four of eight otherwise identical runs of Llama-3.2-3B end with held-out perplexity 5 to 15% above the uncompressed baseline while the other four match it. The effect is in the dispersion, not the mean: the difference in means is +0.58 perplexity at $p = 0.10$, while the variance ratio reaches exact permutation $p = 0.03$. Which runs fail is not predictable from the seed: re-running a damaged configuration produced a safe model twice out of two, and the divergence between two launches disappears under PyTorch’s deterministic kernels. Nine measurements, among them task accuracy, training loss, in-distribution perplexity, gradient norms, four multiple-choice benchmarks over 21 adapters and MMLU, fail to say which run failed. Across 30 compressed runs, dispersion around each arm’s median falls as the gradient cosine of that arm’s backward pass rises ($\rho = -0.62$, $p = 4 \times 10^{-5}$, with no threshold in the statistic); the same test on bits per element gives $p = 0.19$. The sensitive tensors are the query and key head views, and only those: gradient cosine falls to 0.36 there and stays above 0.99 on the value views, the residual stream, the MLP intermediates and the attention output. A five-minute gradient probe identifies them before any training run. Storing those two tensors in eight-bit floating point removes the failure in eight of eight runs, at a cost of 0.60 bits per element at 3B (0.44 at 70B). Scaling them along the channel axis instead costs no extra bits, but leaves one run of eight above the threshold.

1 Introduction

Fine-tuning a 70B model with LoRA on a single device is a memory problem, and after the weights are quantized and the optimizer state is reduced to a few million adapter parameters, what remains

*Code: <https://github.com/yedamkim/actcom>

is activations. On an NVIDIA GB10 with 128 GB of unified memory, Llama-3.1-70B with NF4 weights holds 39.65 GB before a single activation is stored. By a sequence length of 2048, the tensors saved for the backward pass have consumed the rest. Compressing those tensors to four bits brings a 3072-token sequence from a projected 130 GB to a measured 91 GB. The values are packed when they enter the saved-tensor buffer and restored when they leave it. A line of work from ActNN onward has shown that models handle this surprisingly well (Chen et al., 2021; Liu et al., 2022).

They do not handle it uniformly, and the question of which tensors can be compressed has been answered in this line of work by measurement rather than by argument. ActNN and GACT estimate each tensor’s sensitivity at runtime, from the variance its quantization adds to the gradient, and allocate bits accordingly. CompAct leaves the attention head views uncompressed, and HyC-LoRA compresses them without treating them as the sensitive case; in both, the choice is an implementation decision rather than a measurement. The variance estimate is a first-order linearization. It therefore cannot see how much a softmax amplifies a small change in its input. GACT’s own results show this limit: with the estimator running at two bits, reported BERT-large MRPC accuracy falls 5.66 points, against 0.30 at four bits. We give a static answer with a mechanism behind it, and we arrive at it through a failure that standard evaluation does not detect.

We fine-tuned Llama-3.2-3B on GSM8K eight times with four-bit compression applied to every stored tensor, varying only the random seed. Four of the eight ended with held-out perplexity on WikiText-2 between 5.3% and 14.7% above the uncompressed baseline, and the other four were within 0.03 of it. The two groups are separated by a gap of 0.83 perplexity (4.4 standard deviations of our protected configuration, as measured at pre-specification). We report this as an observation, not as a formal test that the runs split into two groups. The variance ratio against the protected arm is 33.7 (exact permutation $p = 0.03$), while the difference in means is +0.58 at $p = 0.10$. At 3,736 steps, compression noise does not degrade the model. It makes the outcome of training unpredictable.

Which runs fail is not predictable from the seed. We re-ran two damaged configurations from scratch, with the same seed and the same settings. Both produced safe models. The loss traces agree bit-for-bit for two optimizer steps, then separate at the third. A probe locates the split: under PyTorch’s deterministic kernels two launches agree bit for bit in every parameter for 20 optimizer steps, and under the default kernels their gradients differ from the first backward pass. Non-deterministic reduction order in the backward kernels is enough to move a run from one mode to the other, so the failure cannot be avoided by choosing a seed.

Task accuracy does not separate them: the four damaged runs average 53.25% and the four safe runs 54.50%, a difference of 1.25 points with an exact two-sided permutation $p = 0.51$ over all 70 splits of the eight observations.¹ Training loss does not separate them either ($p = 0.71$), and neither does perplexity on held-out data from the training distribution. That perplexity varies by only 0.004 across the eight runs, while out-of-distribution perplexity varies by 2.65. Gradient norms show which precision format is in use (586 for uniform INT4, versus 1.2 for the non-uniform grid). But within a single format, the damaged and safe runs are completely mixed together. One run makes the point. The worst of the eight on WikiText-2, at 18.43 against a baseline of 16.08, sits at the median of its arm on in-distribution perplexity and within one standard deviation of its arm’s mean on the task.

This affects how the field evaluates compression. Prior work establishes losslessness through a threshold on in-distribution task metrics (0.5% for ActNN, 1% for GACT). Those metrics do catch catastrophic failures: uniform INT4 costs Adacc (Chen et al., 2025) up to 39% of its accuracy, and

¹Mann–Whitney with mid-rank ties gives the same value.

compressing softmax activations collapses GSM8K accuracy to 0.2%. But they do not catch the near-baseline range in which competing methods are actually compared. Reported gaps between compression variants there are typically under two points, sometimes even favor the compressed model over the uncompressed baseline, and come from a single run. We are not aware of prior work that reports seed variance on held-out perplexity (§2.3), and the failure we describe is invisible without it.

The sensitive tensors are the query and key head views, and only those. We applied four-bit compression to one tensor role at a time and measured the resulting gradient error. The query and key views fall to a cosine of 0.36 against the uncompressed backward pass. The value views stay at 0.99, and so do the residual stream, the MLP intermediates, and the attention output. Yet the value views have the same rank and the same shape as the query and key views, and come from an adjacent projection of the same input. Query and key enter the attention scores as a bilinear product, which is then exponentiated, so a perturbation in either one gets amplified twice. Everything else passes its perturbation through linearly.

The identification is predictive, not just descriptive. If two quantization grids differ only on the query and key views, then protecting those views should make the choice of grid irrelevant and it does. Crossing the body encoding with the treatment of four-dimensional tensors gives a pure interaction: the body encoding moves in-distribution perplexity by 0.077 when the head views are at four bits and by 0.002 when they are at FP8 (three to eight seeds per cell). A five-minute gradient probe predicted a difference that takes ten hours per configuration to measure.

Route the head views to a protected encoding and leave everything else at four bits. Eight-bit floating point removes the failure in all eight runs at a cost of 0.60 bits per element overall at 3B (0.44 at 70B). Computing the four-bit scale along the channel axis rather than within blocks of the head dimension achieves the same variance at no bit cost and no requirement for an eight-bit format, at a mean penalty of 1.7% perplexity. Both work because both raise the same quantity: the gradient fidelity of two tensors that make up an eighth of what is stored.

The measurement that motivates all of this is in §4.2: Llama-3.1-70B LoRA fine-tuning at 3072 tokens fits on one 128 GB device under four-bit activation compression and, by extrapolation, does not fit without it unless activations are recomputed. We report it as context rather than as a contribution. It is one run per length, and §4.3 shows gradient checkpointing reaching an even lower peak. What follows is about the check that such a configuration would have to pass.

- A static rule for which activations handle four-bit compression in a decoder-only transformer. We give the mechanism behind it, a counterexample that rules out tensor rank as the criterion, and a 2×2 design that tests the rule against training outcomes (§5.1).
- A characterization of the failure it prevents: not predictable from the seed, expressed in variance rather than in the mean, and present on two of three held-out corpora (§5.3, §5.4).
- Evidence that the standard evaluation for this class of method cannot detect that failure, across nine measurements taken during and after training, four in-training signals, four multiple-choice benchmarks over 21 adapters, and MMLU over the blockwise arm (§5.5, §5.7).
- A measurement of what the five-minute probe does *not* determine. Two configurations that it scores identically end in measurably different subspaces. We pre-specified the opposite prediction before running the measurement, and we were wrong (§5.1).

2 Related Work

2.1 Compressing the activation store

Backpropagation stores activations, and that store dominates at long context: Wang et al. (2025) measure it at 93.3% of allocated memory for Llama-2-7B under QLoRA at $s=8192$, and Shi et al. (2025) report 64 GB of activations against under 1 GB of gradients and optimizer state. LoRA (Hu et al., 2022) and QLoRA (Detrmers et al., 2023) reduce trainable parameters and weight memory and leave that store untouched. Gradient checkpointing (Chen et al., 2016) removes it by recomputing, at a cost Chen et al. (2025) measure at over 28% of training time; §4.3 compares the two directly.

Activation-compressed training began on convolutional networks: ActNN (Chen et al., 2021) quantizes to 2 bits with per-group scaling under an analytically derived per-operator sensitivity. GACT (Liu et al., 2022) estimates that sensitivity numerically through PyTorch’s `pack_hook/unpack_hook` interface, and AC-GC (Evans and Aamodt, 2021) bounds the expected loss increase. Language models break naive schemes: Chen et al. (2025) report up to 39% accuracy loss or gradient overflow under per-token INT4 and fix it by extracting outlier channels, and Jetfire (Xi et al., 2024) shows why per-token quantization fails (the outliers run along the channel axis) and quantizes per block instead. HyC-LoRA (Wang et al., 2025) is closest to this work: it targets QLoRA fine-tuning with outlier-aware per-channel scaling and compresses nonlinear activations too, because they are 69.7% of the store and cost 10.62 points at 2 bits against 0.99 for linear ones. A parallel line of work stores a low-rank representation instead. CompAct (Shamshoum et al., 2025) under a resampled random projection, LoRAct (Shi et al., 2025) under an online orthogonal decomposition, CARE-LoRA (Zhang et al., 2026) by exploiting that $\nabla_B L$ needs only XA . Wei et al. (2026) give the theory: a zero-mean perturbation leaves linear operators unbiased while nonlinear ones are biased at second order, and the input-gradient error propagates as a product of Jacobians that compounds with depth. Their empirical split is sharp: compressing linear operators alone leaves Qwen3-4B accuracy essentially unchanged, SiLU alone drops it to 0.071 and softmax alone to 0.002.

Work on activation outliers supplies the structural observation this paper builds on. LLM.int8() (Detrmers et al., 2022) keeps outlier channels in higher precision, SmoothQuant (Xiao et al., 2023) shifts scale into the weights instead, and Bondarenko et al. (2023) traces the outliers to specific attention patterns arising consistently across architectures. That work is about inference. What carries over to training is this: what must be protected is fixed by its position in the computation, not by its magnitude. §5.1 makes the claim precise for training and §5.2 shows magnitude failing as a criterion even inside a single tensor.

2.2 Which tensors do prior methods actually compress?

The compression methods surveyed above use similar techniques. What differs is the set of tensors they apply them to. Since our contribution concerns *which* saved tensors handle low precision, we examined the released implementations directly.

ActNN cannot reach attention tensors by design. Its `layers.py` supports `QConvkd`, `QLinear`, `QConvTransposekd`, `QBatchNormkd`, `QSyncBatchNorm`, `QReLU`, `QDropout`, and pooling. There is no quantized `matmul`, `bmm`, `softmax`, or `LayerNorm` layer. Its analytic sensitivity derivations also cover only convolution, linear, and normalization: the activation and pooling wrappers call a fixed kernel and carry no sensitivity scheme. The QK^T product never enters a compression path.

Table 1: Treatment of attention head-view tensors in prior activation-compression methods, determined from released implementations and reported accounting.

Method	Head views	Basis	Precision assignment
ActNN	unsupported	no matmul/softmax layer	per-layer, analytic
GACT	compressed, unaware	rank 2–4 admitted; flat reshape	per-tensor, runtime estimate
CompAct	excluded	Appendix B.2, FlashAttention	rank- r projection
HyC-LoRA	compressed	per-channel; pre-RoPE for Q, K	calibrated, static
Ours (DCR)	compressed at FP8	softmax input path	static, from tensor shape

GACT does compress attention tensors, but without recognizing them as attention tensors. Its `check_quantize` admits any floating-point, gradient-carrying, non-parameter tensor of rank 2, 3, or 4, so head-view tensors of shape (B, H, L, d_h) pass the filter. The quantizer then applies `input.reshape(-1, group_size)` with `group_size = 256`, a flat reshape that respects neither head nor sequence boundaries. GACT does discuss Q, K, V , but only to deduplicate the *shared input* tensor from which the three projections are computed.

CompAct excludes attention tensors deliberately. It lists $q, k, v \in \mathbb{R}^{b \times h \times l \times n/h}$ among the tensors saved for backward, yet they are not projected; only linear-layer inputs are. It explains that the attention output projection was left uncompressed because that activation is shared with FlashAttention, so compressing it saves no memory; a second reason it gives is that doing so would introduce error into the FlashAttention input gradient and risk accumulation across layers.

HyC-LoRA respects head boundaries as a side effect. Because the channel index $c \in [0, d)$ maps one-to-one to (head, head-dim), a per-channel scale can never span two heads. Head structure is not, however, a design consideration: the accounting in its paper treats Q, K, V as $3 \times (b, s, d)$ tensors, and the attention probability map (b, h, s, s) is excluded from quantization and handled by pruning or by FlashAttention. HyC-LoRA does apply special treatment to Q and K : it quantizes them pre-RoPE and re-applies the rotation after dequantization. The stated reason is only to avoid corrupting the data distribution; they attribute the outlier pattern in the Q and K buffers to RoPE itself. Nothing in the paper connects the difficulty of these tensors to the softmax. Their own ablation is consistent with a residual difficulty that per-channel extraction does not reach: raising the channel-extraction ratio sharply reduces quantization loss for RMSNorm and barely moves the curves for SiLU, Hadamard and attention Q alike. The observation is that RMSNorm is the operator this treatment helps, not that Q is singled out; and applying outlier extraction to all operators performs *worse* than their selective scheme.

We summarize in Table 1. Our position is not that attention tensors have gone unnoticed, but that the reason they are difficult has not been isolated: prior work avoids them (CompAct), compresses them without distinguishing them (GACT), or attributes their difficulty to a different cause (HyC-LoRA). Section 5.1 identifies the mechanism directly.

2.3 How is harm measured?

The second question prior work leaves open is what a compression method is checked against. ActNN establishes losslessness through a 0.5% threshold on task accuracy (Chen et al., 2021) and GACT flags any accuracy drop above 1% in its results table (Liu et al., 2022). CompAct reports perplexity for pretraining but evaluates fine-tuning on GLUE (Shamshoum et al., 2025). HyC-LoRA reports

perplexity on WikiText-2, proof-pile and PG19, each from a single run; its only repeated experiment is GLUE, where accuracy is averaged over three runs (Wang et al., 2025). LoRAct fine-tunes on WikiText and evaluates perplexity on WikiText, so its perplexity measurement is in-distribution (Shi et al., 2025). Adacc reports validation loss and zero-shot task scores without perplexity (Chen et al., 2025). CARE-LoRA runs three seeds but does not report perplexity (Zhang et al., 2026).

The metric is almost always in-distribution, and where a held-out corpus is used it is measured once. We are not aware of a prior activation-compression paper that reports the seed-to-seed variance of held-out perplexity. Section 5.3 reports what that measurement finds: a change that moves no in-distribution number and no accuracy threshold, and that half the runs come through unharmed.

3 Method

The method has one main part and several supporting details. The main part is that the tensors entering the softmax bilinear product are stored at a higher precision than everything else; the rest is what makes a measurement of that choice meaningful. Figure 1 summarizes both when compression happens and what it dispatches on.

3.1 Backward-only interception and the quantizer

We intercept the tensors autograd retains through PyTorch’s `saved_tensors_hooks`, so that `pack :  $t \mapsto \tilde{t}$` and `unpack :  $\tilde{t} \mapsto \hat{t} \approx t$` . The forward computation consumes t , not $\hat{t}$, and only $\tilde{t}$ stays in memory between the passes: forward numerics are unchanged by the compression, which we verify bit-for-bit at every configuration. The gradient is not exact, since backpropagation computes with $\hat{t}$, and which tensors can absorb that error is the subject of §5.1. Prior activation-compressed training takes the same backward-only form (Chen et al., 2021; Liu et al., 2022; Wang et al., 2025). The alternative, a straight-through estimator on a quantized forward activation, is quantization-aware training and saves nothing here, because the construction keeps the original tensor in the graph.

The quantizer is symmetric INT4 with a per-block absmax scale over $g = 128$ elements along the trailing axis, two codes per byte, giving $4 + 16/g = 4.125$ bits per element. INT4 rather than FP4 (E2M1) because the per-role probe of §5.1 shows the two differ only on tensors we do not store at four bits anyway, and §5.3 finds no difference between them on any perplexity measurement; the one measurement on which they do differ is the endpoint geometry of §5.1, and we have not measured whether that difference has a downstream cost. Appendix A gives the packing layout, the padding rule and the dequantization path. The eight-seed arms of §5.3 use the E2M1 body encoding, which is also the released code’s default; the INT4 body is reported there as the simpler equivalent on three seeds.

3.2 Protecting the head views

Tensors of rank four are stored differently. In a decoder-only block with FlashAttention the $O(L^2)$ attention matrix is never materialized, so the four-dimensional tensors retained for backward are the attention head views (B, H, L, d_h) together with the rotary embedding tables. Of these, the query and key views are the ones that cannot handle four bits. Compressing them drops gradient cosine to 0.36 against 0.99 for the value views (§5.1). The rule we apply is the following.² It over-selects: of

²The implementation tests $\text{rank}(t) = 4$. Instrumenting the dispatch on Llama-3.2-3B, all 1,064 tensors that pass the filters over two optimizer steps are of rank 2, 3 or 4 with 224, 392 and 448 respectively and none is of higher rank, so the

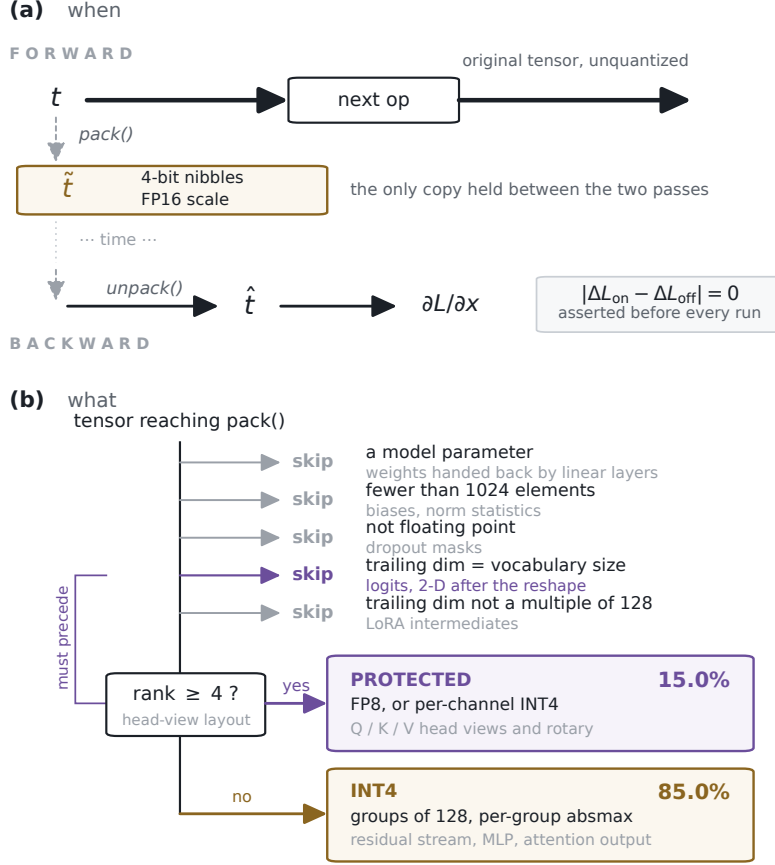


Figure 1: Backward-only activation compression with rank-based dispatch. **(a)** The forward pass uses the original tensor t . Only the packed copy $\tilde{t}$ stays in memory between the two passes, and backpropagation computes with the dequantized $\hat{t}$. So forward numerics stay unchanged. Every run asserts this before training begins: the loss is computed with hooks disabled and enabled, and the run aborts unless the two agree exactly. **(b)** Filters apply in the order shown. The vocabulary-size filter must come before the rank dispatch: after the cross-entropy reshape the logits are two-dimensional and aligned on the trailing dimension, so if a rank rule ran first, it would route them into four-bit quantization by mistake. The two percentages are shares of stored activation elements at Llama-3.2-3B, measured from pack-time counters over a full training run; at Llama-3.1-70B the four-dimensional share is 11.1%.

the 224 rank-4 tensors a step retains at 3B, 112 are rotary tables and the other 112 are head views, 56 query and 56 key or value, so rotary is half the population by count and 5.9% of it by element.

$$\tilde{t} = \begin{cases} Q_{\text{protected}}(t) & \text{if } \text{rank}(t) \geq 4 \\ Q_{\text{INT4}}(t) & \text{otherwise.} \end{cases} \quad (1)$$

FP8 (E4M3) at blockwise granularity costs 8.125 bits per element on the affected tensors and two forms coincide here. We state the rule as ≥ 4 because the property that matters is the head-view layout rather than the exact rank.

produced no damaged run in eight seeds. Symmetric INT4 with the scale computed along the channel axis rather than within blocks of the head dimension stays at four bits, 4.105 bits per element at our sequence lengths, since one scale per channel costs $16/L$ rather than the $16/g = 0.125$ of one scale per group and produced one damaged run in eight, at a mean cost of 1.7% in held-out perplexity. We use FP8 by default and report per-channel as the option when an eight-bit float format is unavailable; §5.3 compares them.

Rank is a stand-in for the property that actually matters. Four-dimensional tensors are not automatically sensitive (ActNN compresses convolutional feature maps to two bits (Chen et al., 2021)). And the rule over-selects: rotary tables are four-dimensional, are re-saved by every layer, make up 5.9% of four-dimensional elements, and are barely affected at four bits. Promoting them wastes 0.035 bits per element, 5.9% of what the promotion costs in total. A rule that inspected the consuming operator would avoid this. We did not implement one, and we note the gap between our rule and our implementation.

3.3 Exclusions and filter order

Not every tensor reaching pack is an activation. The following filters apply in order, and the order matters:

1. **Parameters.** We record the storage pointers of all model parameters before training and skip any tensor sharing one. Without this, `nn.Linear` backward passes the weight matrix to `pack` alongside its input, and quantizing it corrupts the weight gradient. GACT applies the same guard (Liu et al., 2022).
2. **Small tensors.** Fewer than 1024 elements, eight groups at $g=128$. The threshold is set rather than derived; we did not tune it.
3. **Non-floating-point.** Dropout masks and index tensors. At our LoRA dropout rate these are 22.7% of the elements reaching pack at $L=4096$ and 32.2% at $L=512$, counting distinct storage and excluding model parameters.
4. **Vocabulary-sized tensors.** Logits of shape $(B \cdot L, |V|)$. This filter must come before the rank dispatch: logits are two-dimensional after the cross-entropy reshape, so a rank-based rule alone would route them into four-bit quantization. GACT likewise assigns 32 bits to the classification head, after observing on an image benchmark that the context tensor saved by the cross-entropy operator is the most sensitive in the network (Liu et al., 2022).
5. **Misaligned tensors.** Those whose trailing dimension is not a multiple of g ; a group would otherwise span a channel boundary. In practice this excludes only the rank- r LoRA intermediates.

Surviving tensors are dispatched by rank as in Equation 1. Under this policy we compress tensors carrying 68.5% of the activation elements reaching pack at $L=4096$ and 55.3% at $L=512$, over the same denominator: distinct storage, parameters excluded. The difference between the two lengths is that the excluded dropout masks and logits are a larger share at short context: 32.2% and 12.3% against 22.7% and 8.7%.

Two library behaviors store activations at float32 in a configuration that is otherwise four-bit. LlamaRMSNorm upcasts its input to compute the variance and retains that float32 intermediate

for backward, at two sites per block; `prepare_model_for_kbit_training` casts every non-Params4bit parameter to float32, and PEFT’s adapter forward then matches its input, at four sites per block. Restoring both to bfloat16 saves 2.82 and 5.72 GB respectively at $B=2$, $L=4096$, two bytes per stored element, as the tensor shapes predict, and costs no time. Appendix A gives the one-at-a-time decomposition. Every configuration in this paper applies both corrections, so they are part of the baseline rather than a result.

A step whose gradient contains a non-finite value is skipped: the optimizer state is left untouched and the scheduler is not advanced. Twenty-one of the 500 optimizer steps in a Qwen2.5-3B run trigger this. We report the guard as part of the configuration rather than as a result. What we can say about it is limited. Turning it off on Qwen2.5-3B, the only model whose counter is non-zero, flags 21 steps in both directions, and the two index lists differ by exactly the running count of skips ($\text{off}[i] - \text{on}[i] = i$ for all 21) so the same points in the data trigger either way and the guard changes only whether such a step is applied. The unguarded run scores 42.4% against 68.2% for the guarded run with the same configuration. The pair is matched on every recorded configuration field and not on code: the two were launched three weeks apart from different working-tree states, so we read the 25.8-point gap as consistent with the guard’s effect rather than as a measurement of it. Guarded compressed runs on this model span 68.2% to 75.6% across the configurations we have, and the unguarded figure lies below all of them. Appendix A lists both runs and what separates them.

4 Setup and Motivation

4.1 Setup

Llama-3.2-3B-Instruct, Llama-3.1-70B-Instruct and Qwen2.5-3B-Instruct, all on a single NVIDIA GB10 with 128 GB of unified memory. Because host processes share that pool we cap the training container at 120 GB and abort above a reserved-memory ceiling of 100 GB (115 GB for the $L=3072$ cells and for the uncompressed slope measurement at $L=1024$ and 1536), and we report `max_memory_reserved` throughout: on a unified-memory device it is the reserved figure, not the allocated one, that approaches the physical limit.

Fine-tuning is GSM8K (Cobbe et al., 2021), rank-16 LoRA on the attention projections, two epochs at 3B (3,736 optimizer steps) and 500 steps at 70B. Accuracy is 8-shot chain-of-thought with greedy decoding under a fixed stop rule; held-out perplexity is 200 WikiText-2 windows of 512 tokens.

Six arms appear throughout the 3B experiments, and Table 2 names them. An arm is fixed by two choices: the *body encoding*, applied to every compressed tensor of rank two and three, and the storage of the four-dimensional *head views*. Under the method names of §3, the uncompressed arm is Standard, the two arms that leave the head views at blockwise four bits are Four-bit, and the three that route them to a protected encoding are DCR (Dimension-Conditional Routing). The uncompressed arm stores no activation at reduced precision and is therefore independent of the body encoding. The rest of the paper refers to arms by the names in the table; the letters appear only where space is short, in Figure 4 and in the appendix tables. The number of seeds per arm varies with what each claim requires and is stated in every table. Appendix A gives the full configuration, the 70B checkpoint, the sequence-length statistics, and the evaluation protocol.

Table 2: The six arms at Llama-3.2-3B. *body* is the encoding applied to tensors of rank two and three; *head views* is the storage of the four-dimensional attention head views. Appendix B gives the configuration triple that identifies each arm in the released records.

	arm	method (§3)	body	head views	seeds
C	the uncompressed arm	Standard	—	—	2
A	the blockwise arm	Four-bit	E2M1	blockwise 4-bit	8
D	the INT4 arm	Four-bit	INT4	blockwise 4-bit	3
F	the per-channel arm	DCR	E2M1	per-channel INT4	8
B	the FP8 arm	DCR	E2M1	FP8 (E4M3)	8
E	the INT4-body variant	DCR	INT4	FP8 (E4M3)	3

Table 3: Peak reserved memory for Llama-3.1-70B LoRA fine-tuning on a single NVIDIA GB10 (128 GB unified memory), batch size 1, synthetic fixed-length sequences. NF4 weights, adapters, and optimizer state occupy 39.65 GB before any activation is retained. Standard LoRA does not compress activations and is independent of the body encoding.

L	Standard	+ recompute	4-bit, Q/K at FP8	4-bit, Q/K per-channel
1024	71.37	44.53	63.23	61.64
1536	86.13	45.77	—	—
2048	abort [†]	48.63	76.85	74.74
3072	130.4 [‡]	52.67	90.95	87.85

[†] Out of memory at $L=2048$: the watchdog aborted on first observing reserved memory above its 100 GB ceiling, at 100.62 GB against a predicted 100.90. Only 41.18 GB was live at that instant, so the reservation reflects allocator behavior under pressure rather than a lower bound on the working set. [‡] Extrapolated from $L=1024$ and $L=1536$.

4.2 Compression is what makes 70B fit without recomputation

This is the motivation for the rest of the paper. On a single 128 GB device, Llama-3.1-70B LoRA fine-tuning without compression or recomputation exceeds our 100 GB operating ceiling at $L=2048$, at a predicted 100.9 GB that the device holds but the ceiling does not, and extrapolates to 130.4 GB at $L=3072$, beyond the device; with compression it runs at $L=3072$ at a measured 90.95 GB. NF4 weights, adapters and optimizer state take 39.65 GB before any activation is stored, so the activation store is what decides the budget. Gradient checkpointing reaches an even lower peak, 52.67 GB, and §4.3 is where the two are compared; the case for compression is that it costs 7% in step time at this scale where recomputation costs 43%.

These are single runs at each length, and one run cannot separate a configuration from its seed. That is a real limitation for a capability claim, and it is also the reason this section is short. What the paper contributes is the check that such numbers would have to pass, not the numbers themselves. Appendix A records the configuration.

Figure 2 plots the same measurements against sequence length. The uncompressed line is fitted to its two uncensored points, and the compressed lines by least squares over their three points; a two-point fit of the FP8 arm through $L=2048$ and 3072 would place its crossing at 5763 instead of 5815. The uncompressed run at $L=2048$, which the watchdog stopped at 100.62 GB against 100.90 predicted, is right-censored and does not constrain the slope. Weights, adapters and optimizer state are a 39.65 GB constant that compression does not touch, which is why the compressed curves

converge on the baseline’s slope rather than on a fraction of its height.

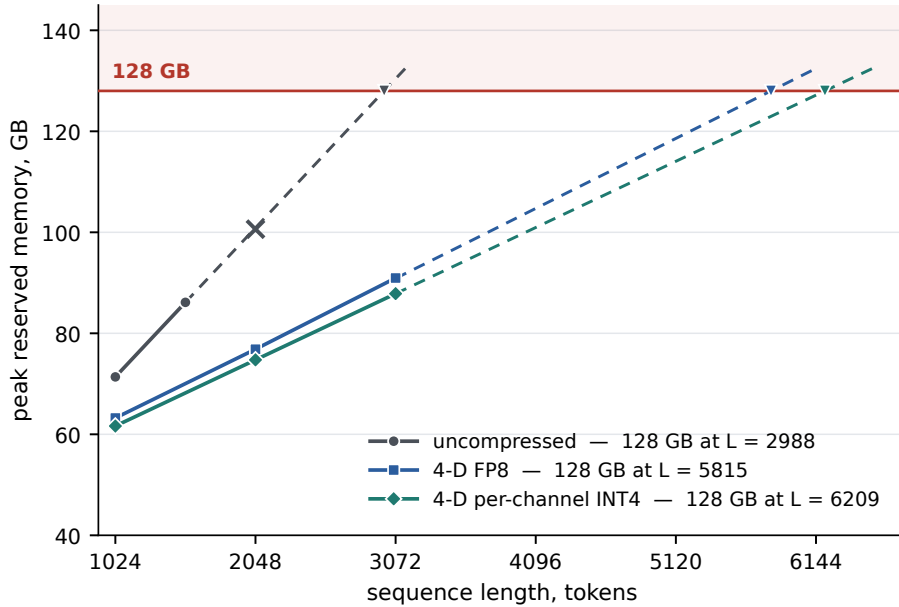


Figure 2: Peak reserved memory against sequence length at Llama-3.1-70B. Round, square and diamond markers are measurements; dashed segments extrapolate to the 128 GB ceiling, where a triangle marks each crossing. The cross at $L=2048$ is the uncompressed run the watchdog stopped.

4.3 The recomputation alternative

Recomputation reaches a lower peak than compression at every length we measured (§4.2), so what justifies compression is time rather than memory. At Llama-3.2-3B, $B=2$, $L=4096$, recomputation reaches $0.43\times$ the uncompressed peak at $1.31\times$ the step time while four-bit compression with FP8 head views reaches $0.70\times$ at $1.18\times$; neither wins on both. The time gap widens with depth: at Llama-3.1-70B and $L=1024$, recomputation costs 43% against compression’s 7%, because recomputation replays a forward pass through 80 transformer blocks rather than 28 while quantization cost scales with stored elements. As a ratio (the only comparison available at $L=3072$, where the uncompressed baseline does not fit), recomputation costs $1.11\times$ compression at 3B, $1.34\times$ at 70B and $L=1024$, and $1.32\times$ at 70B and $L=3072$. Appendix A gives the full timing tables, the two protected encodings’ timings (42.76 against 42.29 seconds at 70B, not separable on one run each), and the extrapolated length at which each budget is used up.

The two can be combined: recomputation discards activations that are never stored, compression reduces those that are, and a configuration that checkpoints some blocks and compresses the rest is available and unexplored here.

5 What Determines Whether Compression Works

5.1 Which tensors are sensitive, and why

Section 3.2 routes four-dimensional tensors to a protected precision without saying which of them needed protecting. This section establishes that only two do, identifies the mechanism, and shows that the identification predicts training outcomes.

Fix a checkpoint and a batch, let $g^\star$ be the gradient with nothing compressed and $g_{\mathcal{F}}$ the gradient with compression applied only to the tensors a filter $\mathcal{F}$ selects, and report $\cos(g_{\mathcal{F}}, g^\star)$ over the concatenated parameter gradients. One role is compressed at a time and everything else passes through untouched, so any drop in the cosine comes from that role alone. Appendix C explains how we attribute tensors to roles: storage-pointer tagging, the re-materialization that rotary embedding and `repeat_kv` perform, and the 6–16% of head views (7 to 18 of the 56 query views) that the tagging misses. It also gives the direction of the resulting bias, which is conservative for the claim below.

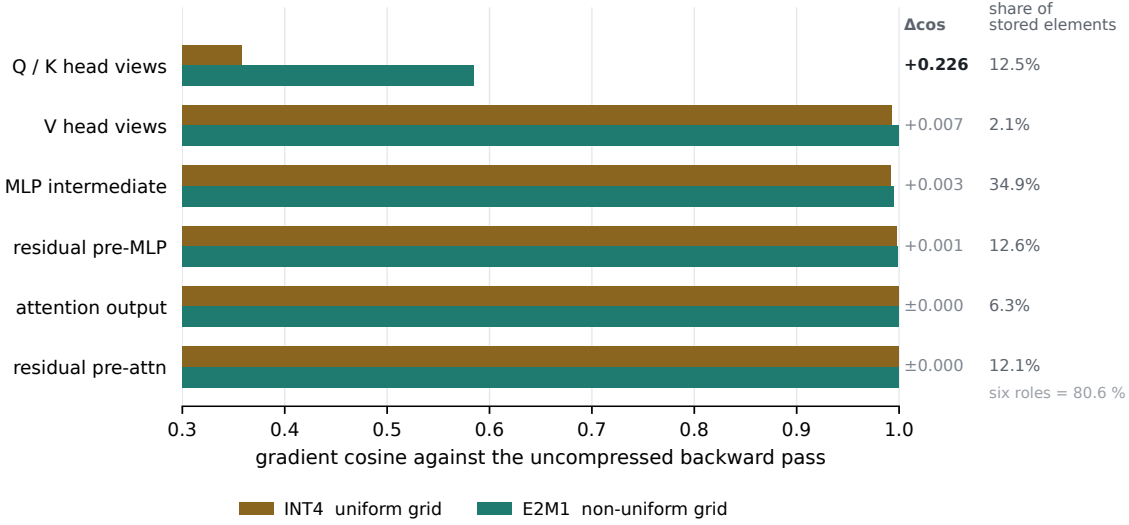


Figure 3: Gradient cosine against the uncompressed backward pass, by role, under the two four-bit grids. Each role is compressed in isolation while every other tensor passes through uncompressed, so any drop in the cosine comes from that role alone. Shares are of the stored elements the probe attributes to a role (Appendix C).

Figure 3 compares uniform INT4 against non-uniform E2M1 at four bits, role by role. The two grids differ by 0.226 in gradient cosine on the query and key head views and by less than 0.01 everywhere else on the residual stream before and after attention, on the MLP intermediates, on the attention output, and on the value head views. Query and key account for 12.5% of the elements attributed in this measurement; the MLP intermediates, at 34.9%, are nearly three times larger and show a difference of 0.003.

The value head views are the informative case. They have the same rank as Q and K , are produced by an adjacent projection of the same input, and differ only in where they enter the attention computation. Compressing them to four bits leaves gradient cosine at 0.992; compressing Q and K leaves it at 0.358.

Q and K enter the attention scores as $S = QK^\top / \sqrt{d_h}$, which is then exponentiated. A perturbation

in either factor gets amplified twice: once by the product, and once by the softmax. The softmax normalizes over the sequence, so shifting one logit relative to its neighbors redistributes probability mass multiplicatively. V is multiplied by the already-normalized weights, and the output projection comes after the attention output. In both, the perturbation passes through linearly.

Wei et al. (2026) derive the general form. For a nonlinear operator the input-gradient error propagates as a product of Jacobians, $\Delta G_X^i = \Delta G_X^l J^{l-1} \dots J^i$, whose Frobenius norms they measure well above 1 at depth; for linear operators the error is unbiased and does not build up. Their experiment shows the same pattern as ours, one level up (operators rather than tensors): compressing linear operators alone leaves Qwen3-4B accuracy unchanged, while compressing softmax alone drops it to 0.002. We locate the effect in the gradients themselves and in a specific pair of tensors.

The rank-4 rule also selects the rotary tables. They are essentially unaffected at four bits ($\cos = 0.9997$, E2M1 grid, step 0), so promoting them costs 0.035 bits per element for no benefit (§3.2).

Table 4: Gradient cosine by depth under four treatments of the query and key head views, at step 0, before fine-tuning. Values are the mean over the LoRA B parameters of each layer: at initialization $B = 0$, so the gradient with respect to A vanishes and the signal is carried entirely by B . The values are from the first of the two probes described above; the arm-level cosines of Table 7 come from its repeat seven minutes later.

Q/K stored as	layer 0	layer 13	layer 27
INT4, blockwise	0.435	0.818	0.815
E2M1, blockwise	0.518	0.849	0.941
INT4, per-channel	0.828	0.974	0.992
FP8	0.984	0.998	0.999

The filters that produce this axis compress the head views and pass everything else through, so two configurations that treat the head views alike receive the same measurement by design, no matter what else differs between them. We tested what that blindness costs when the probe is used to predict. The FP8 arm and its INT4-body variant (Table 2) route the head views to FP8 and differ only in the body encoding, E2M1 against INT4. The probe therefore gives both a cosine of 0.9919 and a norm ratio of 0.9518. Before measuring, we pre-specified the prediction that their training endpoints would be indistinguishable in geometry as well.

They were not. The mean cosine between the row spaces of two runs’ LoRA updates is 0.1339 in the FP8 arm and 0.1314 in the INT4-body variant, and the two arms’ pair values do not overlap. Sample size does not explain the gap: a variance-component model over the eight-run arms puts it at $z = 5.9$ to 7.1 . The metric does not explain it either: row-space angles are exactly scale-invariant, and the divergence is in the largest singular directions, not the smallest. The body encoding has no visible effect anywhere the probe and the perplexity measurements look: every body role stays above $\cos = 0.99$ under either grid, and the INT4-body variant’s mean falls inside the FP8 arm’s range on all four held-out corpora. It does have an effect at the endpoint: the INT4-body variant’s adapters finish with a mean update norm of 24.5 against 31.0 for the FP8 arm, which is the level of the uncompressed runs.

We report this as a limit on what the probe can tell us. It orders configurations by how faithfully the backward pass survives compression of the tensors it compresses. It does not determine where training ends up, because an axis it does not measure moved the endpoint while leaving both of its numbers, and four corpora, unchanged.

Under blockwise four-bit compression the first transformer layer is consistently the most sensitive, its gradient cosine falling 0.31–0.42 below the deeper layers across two probes run seven minutes apart. That spread narrows to 0.15 under a per-channel scale and to 0.014 under FP8, tracking the aggregate cosine. Layer 0 is the only depth whose cosine moves noticeably between the two probes by up to 0.026, against 0.011 at layers 13 and 27, so we report its gap as a range rather than a point. We do not know why the first layer is worse. A per-layer routing rule that protects only the earliest blocks is a natural extension we did not try.

If the two grids differ only on Q and K , then protecting Q and K should make the choice of grid irrelevant. Table 5 tests this directly by crossing the body encoding with the treatment of four-dimensional tensors.

Table 5: WikiText-2 perplexity crossing the body encoding with the treatment of four-dimensional tensors. Each cell is one seed (42) at 3B.

	4-D at four bits	4-D at FP8	4-D effect
INT4 body	17.694	15.925	+1.769
E2M1 body	15.780	15.933	−0.153
body effect	+1.914	−0.008	

The body encoding is worth 1.91 perplexity points when the head views are left at four bits and −0.008 when they are not; the effect vanishes. Seen from the other side, upgrading the head views is worth 1.77 points under an INT4 body and −0.15 under E2M1. There is no main effect, only an interaction, and it is the interaction the per-role probe predicts.

Table 5 is one seed per cell, and seed 42 is the lowest of the eight blockwise runs on WikiText-2, so its cell values are not stable on their own. The same 2×2 on in-distribution GSM8K perplexity, with three to eight seeds per cell, has the same shape: the body encoding moves it by 0.077 when the head views are at four bits (2.6415 against 2.5645) and by 0.002 when they are at FP8 (2.5592 against 2.5577), and upgrading the head views moves it by 0.082 under an INT4 body and 0.007 under E2M1. With head views at FP8, an INT4 body reaches in-distribution perplexity 2.5592 across three seeds (sd 0.0013) against 2.5580 for the uncompressed baseline (two seeds, sd 0.0009), and a gradient norm of 0.62 against 0.61 for an E2M1 body and 0.59 uncompressed, while the same INT4 body with head views at four bits reaches 2.6415 and 586.

5.2 Within a tensor, magnitude does not identify what to protect

The method of §3 allocates precision *between* tensors. The within-tensor approach is the more obvious one, and it is the one we built first. It fails.

Quantization error under a per-group absolute-maximum scale is bounded by the group’s maximum: a group whose largest element is m has step size $m/7$ at four bits, so the groups holding the largest values contribute most to the error in absolute terms. Scoring groups by max-abs and promoting the top $K\%$ to FP8 targets the error where it is largest, at $4K/100$ bits per element. This is the within-tensor version of outlier-aware weight quantization (Dettmers et al., 2022), and HyC-LoRA applies a related idea to buffered activations (Wang et al., 2025).

Max-abs ranks groups by how badly the quantizer reconstructs them. Whether a group’s error matters is a different question, and the two rankings are almost unrelated. Over 15 checkpoints of

a 300-step run we ranked the 128-element groups of each residual hidden state by their maximum absolute value and, separately, by the norm of the gradient flowing through them. The Spearman correlation between the two rankings is 0.13. The overlap between the top 20% under each is 0.248, against 0.200 expected by chance: selecting on max-abs is barely better than selecting at random.³

Section 5.1 explains why the two rankings disagree. The residual stream carries activations whose dynamic range, absolute maximum over median absolute value, is 9362 at its largest against 125 for the query views, a factor of 75 (the largest per-tensor ratio in the per-role probe of §5.1: step 0, before fine-tuning, $B=1$, $L=512$), and yet it is entirely insensitive to four-bit compression while the query views are not. A statistic that measures how extreme a tensor’s values are is measuring the wrong thing.

We compared $K=0$ against $K>0$ on three model–scale combinations. No comparison favors the anchors. One of the four, the Qwen pair with an E2M1 body, reaches nominal significance (McNemar exact $p = 0.037$), in the direction against the anchors; the other three do not (Table 6).

Table 6: GSM8K accuracy with anchors against without, in percentage points. Δ is the anchored arm minus the unanchored one, so a negative value means the anchors did not help. Every comparison is negative except the 70B pair, whose difference is well inside its own standard error. The 3B rows use the in-training evaluation (500 problems, 256-token budget, no stop sequences), the 70B row the unified re-evaluation of Table 14 (200 problems, 512 tokens, stop sequences), and the Qwen rows a 500-problem re-evaluation at 512 tokens without stop sequences; comparisons are made within a row only.

Model	Comparison	Δ	
Llama-3.2-3B	$K=0.2$ vs. $K=0$, paired	−1.40	$t = -1.15$, $df = 2$
Llama-3.1-70B	$K=0.2$ vs. $K=0$	+0.50	$n=1$ each, difference SE 3.2
Qwen2.5-3B	$K=0.2$ vs. $K=0$, INT4 body	−2.60	McNemar exact $p = 0.24$
	$K=0.2$ vs. $K=0$, E2M1 body	−3.80	McNemar exact $p = 0.037$

The 3B comparison pairs by seed and is limited to the three seeds for which both arms were evaluated under the same protocol and neither run is a duplicate training (§5.4).⁴ The Qwen comparisons are paired by question. All arms answer the same 500 items, so McNemar’s test applies where per-item records exist; the two INT4-body arms disagree on 103 of the 500 items with a net difference of 13, and the two E2M1-body arms on 75 with a net difference of 19. We report the direction because it is consistent across the four comparisons in the table, and we do not claim it is established; the excluded pair goes the other way.

They do have an effect. On Qwen2.5-3B, whose activation outliers are an order of magnitude larger than Llama’s, the number of optimizer steps skipped for non-finite gradients falls from 21 at $K=0$ to zero at $K=0.4$: promoting the largest groups does suppress the overflow that the safeguard of §3.3 otherwise catches. The mechanism works, but it is redundant. The safeguard already handles those steps at no cost in bits, and the precision the anchors consume buys nothing.

³Llama-3.2-3B, BF16 base weights, $B=1$, $L=512$, measured every 20 steps with three samples per point; this predates the E2M1 body encoding and the rank-4 routing rule. The result file, `results/gradient_correlation_20260329_000456.json`, also predates the released development history, and the released code contains no script that regenerates it; the file is the only record of this measurement.

⁴A fourth pair exists at seed 789 under the same in-training protocol, 56.6% with anchors against 54.0% without, in the direction favoring the anchors. Its unanchored member is the first of the two trainings of the FP8 arm at that seed, which the rule of §5.4 excludes from every aggregate. Including the pair would move the 3B Δ from −1.40 to −0.40.

The two ways of allocating precision are not interchangeable. Within a tensor, the largest values are the ones a max-abs scale already handles best in relative terms. Between tensors, the question is which tensors sit on a path that amplifies error, and §5.1 shows that question has an answer.

5.3 How much precision the head views need

Section 5.1 established that the query and key views are the sensitive tensors. This section asks how sensitive: we vary the precision at which they are stored across four settings, hold everything else fixed, and measure held-out perplexity over eight seeds per setting.

The four settings differ only in how the head views are quantized, and are identified by the gradient cosine each produces on a fixed checkpoint (§5.1):

Table 7: The four points of the dose–response curve. Cosines come from the per-role probe of §5.1 and share its attribution gap: each is measured with the query views the pointer map misses left uncompressed, which biases all four in the same direction and toward one, so the ordering is unaffected and the absolute values are conservative. Q views is how many of the 56 query head views the filter reached in that measurement (Appendix C); the FP8 point is the least covered, so its cosine carries the largest share of that bias. The bits column is the cost on the head-view tensors themselves; over the whole activation store the per-channel arm measures 4.1205 bits per element against 4.125 for blockwise and 4.727 for FP8.

Storage	$\cos(g, g^\star)$	Q views	bits/elem	how it arises
INT4, blockwise (D)	0.371	47/56	4.125	INT4 body, no rank-4 rule
E2M1, blockwise (A)	0.587	48/56	4.125	E2M1 body, no rank-4 rule
INT4, per-channel scale (F)	0.910	43/56	4.105	rank-4 rule, four-bit
FP8 (E4M3), blockwise (B, E)	0.992	40/56	8.125	rank-4 rule, eight-bit

The first two come from letting the body encoding handle the head views; the gradient measurement in §5.1 showed that INT4 and E2M1 differ only on these tensors, and here that difference becomes the first two points of a dose–response curve. The last two are the two protected encodings of §3.2.

Figure 4 plots every run. The pre-specified threshold is $\tau = 16.65$: the uncompressed mean, 16.08, plus three standard deviations of the FP8 arm as it was at pre-specification, 0.19 over its first three seeds, fixed before any of the additional seeds were run. Under it the damage counts are 3/3, 4/8, 1/8, and 0/8 as gradient cosine rises. Both spread and mean move monotonically (Table 8).

Table 8: WikiText-2 perplexity by gradient cosine, aggregated over the runs of each arm, with each arm’s comparison against the FP8 arm in the three right-hand columns. *range* is the largest value minus the smallest, *damaged* counts runs above the pre-specified threshold $\tau = 16.65$, and *gap* is the largest gap between adjacent runs. The uncompressed row is the reference the threshold was built from.

Storage	cos	n	mean	sd	range	damaged	$\Delta(t, p)$	ratio(p)	gap
INT4, blockwise (D)	0.371	3	22.90	9.395	16.48	3/3	—	—	16.05
E2M1, blockwise (A)	0.587	8	16.66	0.876	2.65	4/8	+0.58 (1.86, 0.10)	33.7 (0.03)	1.41
INT4, per-channel (F)	0.910	8	16.35	0.233	0.67	1/8	+0.28 (2.84, 0.015)	2.37 (0.20)	0.16
FP8, E4M3 (B)	0.992	8	16.07	0.151	0.41	0/8	—	1	0.19
uncompressed (C)	—	2	16.08	0.007	0.01	0/2	—	—	0.01

A damage count depends on where the line is drawn, and §6 records how much. The trend

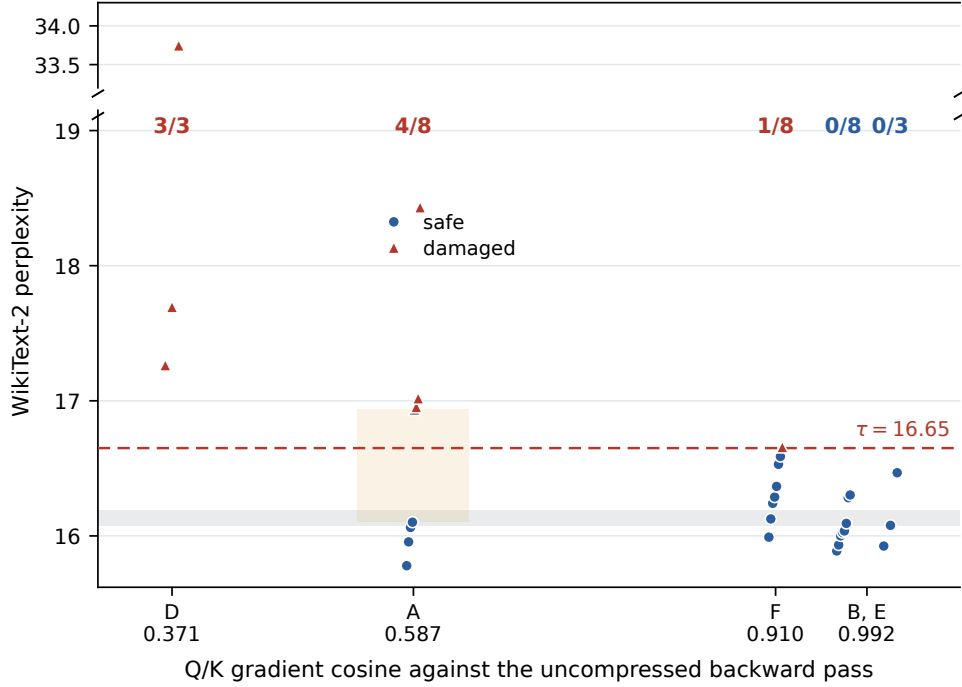


Figure 4: WikiText-2 perplexity of every 3B run against the Q/K gradient cosine of its configuration, one mark per run. The dashed red line is the pre-specified threshold $\tau = 16.65$; the counts above each column are runs above it. The gray band spans the uncompressed adapter (16.08) to the un-fine-tuned base model (16.19), and the pale band at A (blockwise) is its empty interval, 0.83 wide. The y axis is broken for the outlier of D (INT4) at 33.75. Ticks name the arms as in Table 2. Left to right the columns are D (INT4), A (blockwise), F (per-channel), B (FP8) and E (INT4-body variant), as defined in Table 2. The two columns at 0.992 share the storage and differ in body, E2M1 ($n = 8$) and INT4 ($n = 3$), and are drawn apart.

can be tested without drawing one. Write x_{ij} for the WikiText-2 perplexity of run i in arm j and $z_{ij} = |x_{ij} - \text{median}(\text{arm } j)|$ for its distance from its own arm’s center, the Brown–Forsythe transform, which turns a question about dispersion into a question about location. The statistic is then the Spearman correlation between an arm’s gradient cosine and the z of its runs, over every compressed run at once.

Counting threshold crossings misses part of what separates these settings, because the two four-bit schemes fail in opposite ways. Blockwise four-bit fails in the variance (Table 8). Its mean differs from the FP8 arm at $p = 0.10$, its variance is 33.7 times larger at $p = 0.03$, and the median-centered Brown–Forsythe statistic agrees, $W = 9.6$ at permutation $p = 0.019$.

We use Welch’s t for the means and a permutation test for the variance ratio. In Table 8, Δ is each arm’s mean difference from the FP8 arm with Welch’s t and its p , and $ratio$ is its variance ratio against the FP8 arm with the exact two-sided permutation p on $|\log F|$ over all relabelings of the two arms; the $n=3$ and $n=2$ rows are left blank on both tests. The F -test’s normality assumption fails at $n = 8$: 59% of this arm’s variance is carried by one run, and that is exactly the tail the F -test takes its p -value from. Appendix F says why swapping the two tests in the other direction does not work either. The conclusion does not depend on the choice: a permutation test of the mean difference gives $p = 0.065$.

The arm also separates into two groups. Four runs fall between 15.78 and 16.10 and four between 16.94 and 18.43, with no run in the 0.83 between them, which is 4.4 times the standard deviation of the FP8 arm at pre-specification, the quantity that defined the threshold. We report this as an observation rather than as a test: with eight points, no test for two groups reaches significance unless it fixes the split position in advance (Appendix F). None of the claims in this paper rests on the separation. The variance ratio, the Brown–Forsythe statistic and the dose response are all computed on the arm as a whole.

Per-channel four-bit fails in the mean, and only slightly. Its variance ratio against FP8 is 2.37 at $p = 0.20$ and its distribution does not split into two groups (Table 8). But it sits 1.7% higher on average, at $p = 0.015$. Its single threshold crossing is a run at +3.6%, below the lowest blockwise failure at +5.3%; the threshold was built to fall inside a gap and here there is no gap for it to fall into, so it cuts off the top of a continuous spread instead.

Table 9: Dispersion against the dose axis, over every compressed run at once. Each run’s z is its distance from its own arm’s median; the statistic is the Spearman correlation between the arm’s score and z . The null reassigns runs to arms of the observed sizes and recomputes every median before recomputing z ; one-sided, 10^6 draws. Preserving the group sizes matters in the three-arm subset, where the $n=3$ arm’s median is one of its own points and so contributes an exact zero to z .

dose axis	runs	ρ	p
gradient cosine	30, five arms	−0.624	3.9×10^{-5}
gradient cosine	27, INT4-body variant dropped	−0.637	1.8×10^{-5}
gradient cosine	19, bits held at 4.105 to 4.125	−0.570	5.5×10^{-5}
bits per element	30, five arms	−0.234	0.19
bits per element	19, bits held at 4.105 to 4.125	+0.701	1.000

Dispersion falls as gradient fidelity rises, with no threshold anywhere in the calculation (Table 9).⁵ The bits axis cannot order these arms: the two most damaged, INT4 and E2M1 blockwise, are tied at 4.125 bits, and per-channel INT4, which is nearly as clean as FP8, is *below* both at 4.105. What predicts the damage is not how much is stored but how faithfully the gradient survives it.

The threshold was fixed from the first three FP8 runs; recomputed from the final eight it is 16.53 (Appendix F). The blockwise count is unchanged, because both values fall inside its empty band, and the per-channel count moves from one to three. Across 2σ , 3σ and 4σ definitions the blockwise arm flags the same four runs, while the per-channel arm gives three, one and zero. Threshold sensitivity tracks the presence of a gap: where the arm has one, the count is robust to where the line is drawn; where it does not, the count depends only on where the line is drawn. This is why we report the pre-specified count and the distribution together.

Compared with prior work, our threshold is strict, not loose. Our threshold sits 3.6% above the uncompressed baseline. Wang et al. (2025) report a 5.6% increase in WikiText-2 perplexity at the two-bit setting their abstract highlights, and 1.1% at four bits, both as successes. The failure we describe therefore lies inside the band that prior work reports as working, which is the same point §2.3 makes about the metrics: the threshold is not loose; it measures something the accepted thresholds do not.

⁵Shuffling the z values instead of reassigning runs carries the original grouping inside the statistic and is conservative by a factor of 3.2: on the same 30 runs and the same draws it gives $p = 1.25 \times 10^{-4}$.

Two settings share the FP8 point on the curve: an E2M1 body ($n=8$) and an INT4 body ($n=3$). They agree on all four corpora, on in-distribution perplexity, and on gradient norm (§5.1), and neither produces a damaged run. Once the head views are protected, the grid used for the remaining 85% of stored elements does not matter on any of the measurements just listed. This is why we report INT4 as the body encoding in §3.1: it is the simpler grid, needing only rounding where E2M1 needs a lookup table, and at this point on the curve the two are indistinguishable on those measurements.⁶ The two grids do differ in endpoint geometry (§5.1), a difference whose downstream cost we have not measured.

Therefore, FP8 is the safe choice: no damaged run in eight seeds, no mean penalty, and a spread of 0.41 across eight runs against 0.01 for two uncompressed runs. It costs 0.60 bits per element overall at 3B, where the head views are 15% of the store (0.44 at 70B, where they are 11%), and requires an eight-bit floating format. Per-channel scaling stays at four bits and needs only nibble packing and no FP8 support, at a mean cost of 1.7% held-out perplexity and one boundary crossing in eight seeds. Blockwise four-bit is not a usable option at either cost. It introduces run-to-run failures, and §5.7 shows that no measurement taken during or after training on the task distribution detects them. Every rate in this section is for 3,736 optimizer steps. At 500 steps no arm is stable on held-out text, the uncompressed arm included (Appendix E), so the FP8 result is a statement about this training length rather than a general one.

5.4 The failure is not predictable from the seed

Section 5.3 reported that four of eight runs under blockwise four-bit compression exceed the damage threshold. This section characterizes those failures: which runs fail is not determined by the seed, the degradation is a change in variance rather than in mean, and it does not look the same on every corpus.

We re-ran two damaged configurations from scratch with the same seed, the same data order and the same settings, and both became safe. The two runs of a pair were launched from different working-tree states of the development repository (commits 6b2aa55 and 77e9837 for seed 123, 7bde870 and 77e9837 for seed 4042, each with uncommitted changes), so we cannot certify the code identical; the commits between them add a memory watchdog, a further body-encoding option and evaluation chains (docs/development_history.txt). The divergence is visible in the loss trace: the first two optimizer steps agree bit-for-bit, which is evidence that the training path was numerically the same, and the third does not (Table 10).

These three pairs are also the only comparison with the seed held fixed for the endpoint geometry later in this section, where they separate the two arms.

We attribute this to non-deterministic reduction order in CUDA kernels, and a probe supports the attribution: with `torch.use_deterministic_algorithms` and a fixed cuBLAS workspace, two launches of the blockwise, FP8 and uncompressed configurations at seed 123 agree bit-for-bit in loss and in every LoRA parameter through 20 optimizer steps, while under default kernels the same pairs separate by step 3 (Table 27, Appendix F). Under the default kernels the two launches already differ in the gradients of the first backward pass; the first optimizer update hides the difference, the second writes it into the parameters, and the loss shows it at step 3, which is the pattern of Table 10. Of the kernels of the training step that we tested, the flash-attention backward is the one that returns different

⁶We make no step-time claim between them: the same configuration measured twice on the same day differs by a factor of 1.47 in seconds per step, so no difference between the two encodings could be seen through that noise.

Table 10: Re-running two damaged configurations with the seed, the data order and the settings unchanged. Upper: WikiText-2 perplexity of the original run against the re-run; both cross back below the damage threshold $\tau = 16.65$. Lower: the training loss at seed 123, which agrees to six decimal places for two optimizer steps and separates at the third.

seed	original	re-run	
123	18.432	16.271	damaged $\rightarrow$ safe
4042	16.935	16.351	damaged $\rightarrow$ safe

seed 123	step 1	step 2	step 3
original	1.295640	1.641697	1.472919
re-run	1.295640	1.641697	1.473335

bits when called twice on the same input, and deterministic mode makes it repeat. Under deterministic kernels the outcome is therefore a function of the seed, but one so sensitive to a one-ulp perturbation of the gradient that the seed does not predict it; a user who fixes the kernels can reproduce a given run, at a step-time cost of 13 to 18%, and still cannot choose a safe one. The same pattern appeared on its own in a third case: an FP8 run at seed 789 was accidentally trained twice and diverged at the same point. Both of those runs were safe, and the FP8 arm shows no such spread (15.999 and 16.093 on WikiText-2 for the duplicate pair).

The consequence is that a user cannot avoid this failure by choosing a seed. Under the default kernels, two runs of the same configuration are two random samples.

The re-runs are excluded from the damage rate. They were selected because their originals were damaged, so counting their outcomes would bias the rate toward whatever the re-runs happened to produce, which here would be downward, since both came back safe. The rate of four in eight is over the original eight runs only, and the re-runs are evidence about what a repeat draw does, not additional samples of the rate.

We have one observation on a second model, and it is confounded. Two Qwen2.5-3B runs whose recorded configurations differ only in output directory, made on the same day, met non-finite gradients at 20 optimizer steps and at none, and scored 40.0% and 70.8% on GSM8K. The record of the first shows that its flagged steps were not skipped the way the guard of §3.3 skips them in every later run, so the scores reflect how those steps were handled as much as the run, and we do not count the pair as a second instance; Appendix A gives the record. What it does show is that whether overflow occurs at all varies between runs of one configuration on that model.

If damage were seed-determined, pairing the four-bit and FP8 runs by seed would cancel the seed effect and shrink the standard error. It does not: the paired t is 1.85 against an unpaired Welch t of 1.86. Pairing recovers nothing, as expected when the pairing variable does not determine the outcome. This matches the re-run result above, where the same seed produced a safe model twice out of two.

Because the draws are independent, the eight runs are an i.i.d. sample and the binomial test is exact: four of eight, against zero of eight under FP8 (one-tailed Fisher exact $p = 0.038$; the direction was fixed in advance by defining the threshold from the FP8 arm; on the 24 runs evaluated after the threshold was fixed the count is 2 of 5 against 0 of 5, one-sided $p = 0.22$).

Across three held-out corpora, blockwise four-bit compression does not shift the mean but does inflate the variance:

Table 11: Mean difference and variance ratio between the blockwise arm and the FP8 arm, eight runs each, on three held-out corpora. Appendix F gives the run-selection rules behind the table.

corpus	mean difference		variance ratio	
	Δ	$t(p)$	F	p
WikiText-2	+0.585	1.86 (0.10)	33.7	0.030
GovReport	+0.103	1.31 (0.23)	14.0	0.029
NarrativeQA	+0.363	0.88 (0.40)	4.7	0.209

Table 11 gives the result. Mean differences are Welch t and variance ratios are exact permutation, for the reasons given above. No mean difference reaches $p = 0.05$. Two of the three variance ratios do: WikiText-2 at 0.030 and GovReport at 0.029, with NarrativeQA at 0.209. The median-centered Brown–Forsythe test agrees on the same two, at 0.019 and 0.014 against 0.229 for NarrativeQA.

Reading the Brown–Forsythe statistics themselves would suggest otherwise, since they are 9.6, 2.9 and 1.8. But a statistic is only large relative to the distribution it came from. Against its own permutation distribution each of the first two sits in the upper tail: 9.6 against a 95th percentile of 8.2 on WikiText-2, and 2.9 against 2.3 on GovReport. NarrativeQA’s distribution is far wider, with a 95th percentile of 10.2, so 1.8 there is nothing special. The three raw statistics are on three different scales, so comparing them directly, as a shared parametric reference would invite, is not meaningful. The two additional corpora were not pre-specified, so we read them as supporting rather than as independent tests.

The dispersion is not measurement noise. Re-evaluating a saved adapter reproduces perplexity to four decimal places: we checked three adapters, including the most extreme, and the difference was ± 0.0000 . The two uncompressed runs differ by 0.009. The 0.88 standard deviation under blockwise four-bit compression is variation between models, not between measurements of the same model.

Perplexity and benchmark accuracy both measure the model through a data distribution. The LoRA update is the model, so we compared the endpoints directly. Write $\Delta W = (\alpha/r)BA$ for the update at one of the 112 adapter sites. We compare the product, since the factors are identified only up to an invertible G acting as (GA, BG^{-1}) . Two runs can then differ in how far apart their updates are, and in the subspaces those updates occupy.

They do not differ in distance. The mean Frobenius distance between two blockwise updates is 38.8 against 37.1 for FP8, a gap of 4.6% at exact permutation $p = 0.25$. They differ in direction. Averaging over the sixteen principal angles between the row spaces of ΔW at each site, two blockwise runs sit at a mean cosine of 0.1121 and two FP8 runs at 0.1339: the blockwise arm occupies more varied subspaces. Extending the measurement to the INT4 and per-channel arms orders all four by gradient cosine without exception, with the uncompressed pair above them all.

The ninety-one within-arm pairs, the eighty-eight of Table 12 and the three of the INT4-body variant, do not overlap between arms at all. Under the null that reassigns the 27 compressed runs to arms of the observed sizes, no reassignment in 10^5 separated four arms this way, and correlating each pair’s value with its arm’s gradient cosine gives $\rho = 0.949$ at $p = 10^{-5}$; scoring by bits per element instead gives 0.447 at $p = 0.047$. We do not quote a rank correlation on the four arm means, which is +1.000 but reaches that value in about one reassignment in six and so carries almost no information at four groups.

Three reference points show what these numbers mean. Two unrelated 16-dimensional subspaces

Table 12: Mean cosine of the principal angles between the row spaces of the LoRA updates, over all within-arm pairs and all 112 adapter sites. Lower means the two runs occupy more varied subspaces. *range* is the smallest and largest pair value in the arm; no two arms overlap. The uncompressed row is a reference point from a single pair, not an estimate.

arm	$\cos(g, g^*)$	pairs	mean	range
INT4, blockwise (D)	0.371	3	0.0829	0.0823–0.0840
E2M1, blockwise (A)	0.587	28	0.1121	0.1113–0.1134
INT4, per-channel (F)	0.910	28	0.1231	0.1218–0.1242
FP8 (B)	0.992	28	0.1339	0.1330–0.1350
uncompressed (C, $n=1$ pair)	—	1	0.1391	—

of $\mathbb{R}^{3072}$ meet at a mean cosine of 0.061; two runs sharing a seed but differing in compression sit at 0.82, since LoRA’s A is initialized from the seed; and the within-arm values above lie between. Reading the arms as a fraction of the interval from chance to the uncompressed pair, INT4 recovers 28% of it, E2M1 blockwise 65%, per-channel 80% and FP8 93%. Every arm sits well above chance, and the uncompressed pair itself sits only 0.078 above it, so the scatter is not something compression creates. Training pulls the endpoints toward a common direction from the start, and compression weakens that pull by a measurable fraction.

The three re-run pairs above are the only place in this work where a run is repeated with the seed held fixed, and the same measurement applies to them. The two blockwise re-runs sit at 0.832 and 0.829, the same distance apart as two runs that differ in compression scheme entirely, while the FP8 re-run sits at 0.954. Re-running the blockwise configuration moves the endpoint about as far as changing the configuration does; re-running FP8 does not. With only two pairs against one, we describe this but do not test it. Controlling the seed also makes the Frobenius distance separate the arms, where between arms it was null: the two blockwise re-runs sit at 28.4 and 32.9 against 21.5 for FP8, a ratio near 1.4, while the between-arm figure was 1.05. At the arm level, the seed variance hid the effect; the effect was there.

What the same-seed pairs do *not* show is where the damage went. The two blockwise pairs differ by a factor of 3.7 in how far their perplexities moved apart, 2.161 against 0.584, and by 0.003 in subspace cosine. Holding the seed fixed makes the amplification visible; it does not make the location of the damage visible.

Table 13 collects the two levels. Between arms, each test is a two-sided permutation over the 12,870 relabelings of the sixteen runs, using the variance ratio for the two run-level quantities (perplexity, and accuracy averaged over the four benchmarks) and the difference in mean within-arm pair value for the three pairwise ones (item disagreement, subspace cosine, Frobenius distance). Within the arm, each test is an exact two-sided Mantel test over the 40,320 relabelings of the eight runs, correlating each pair’s value with the pair’s absolute difference in held-out perplexity (for accuracy, the pair’s value is the absolute difference in mean accuracy). The first column is full of significant results; the second holds one signal that is significant on paper (subspace, $p = 0.025$) that does not survive leave-one-out (3 of 8). So no metric aligns with damage at the run level, and this is the paper’s result, not a gap in it.

Compression widens the spread of where training ends up, in weight space, in held-out perplexity, and in item-level behavior, at the level of the configuration. The three do not agree run by run: the one

Table 13: Every dispersion measurement in this paper, tested at two levels. *Between arms* asks whether the blockwise arm is more dispersed than the FP8 arm; *within the arm* asks whether the measurement identifies which run inside the blockwise arm is the damaged one. The parentheses on the accuracy and item-level rows repeat the within-arm test on MMLU’s 14,042 items; the between-arm tests cannot be repeated there, because only the blockwise arm was evaluated on MMLU (§5.5).

measurement	between arms	within the arm
held-out perplexity	$p = 0.03$	by construction [†]
item-level behavior	$p = 0.001$	$p = 0.21$ (MMLU 0.67)
LoRA subspace direction	$p = 0.0002$, no overlap in 91 pairs	$p = 0.025$ [‡]
LoRA update distance	$p = 0.25$	$p = 0.22$
benchmark accuracy	$p = 0.31$	$p = 0.93$ (MMLU 0.35)

[†] The damage label is defined by a threshold on held-out perplexity, so perplexity separates damaged from safe runs by definition and not as a finding. It is listed to make the circularity explicit.

[‡] Leave-one-out over the eight blockwise runs: p stays below 0.05 in three of the eight drops (0.004, 0.027, 0.041) and lies between 0.10 and 0.21 in the other five.

run-level signal, in subspace direction, is nominally significant and does not survive leave-one-out. That is what “each run reaches a different minimum” means in practice, and it is why no single downstream measurement detects the effect: a run that landed somewhere else is worse on some distributions and better on others, and which distributions those are changes from run to run.

For example, seed 42 of the blockwise arm scores 15.78 on WikiText-2, better than both uncompressed baselines (16.07, 16.08) and better than the base model before fine-tuning (16.19).⁷ Seed 123 scores 18.43. Compression noise perturbs the training trajectory; it does not bias it.

The four runs flagged on WikiText-2 are not the four runs flagged elsewhere. The threshold formula is the uncompressed mean plus three standard deviations of the FP8 arm, taken over all eight FP8 runs since nothing was fixed in advance there; with the three-seed standard deviation of the pre-specified rule it gives 3/8 and 2/8. Applied after the fact to the other two corpora it gives 2/8 on NarrativeQA and 1/8 on GovReport, and the sets overlap in one seed each. Rank agreement between corpora is near zero for the most distant pair (WikiText-2 against NarrativeQA, Kendall $\tau = -0.07$, $p = 0.91$).

We do not count these crossings toward the damage rate; three corpora with independently applied thresholds would inflate it, and only WikiText-2 was pre-specified. What they show is that the perturbation does not push every run in the same direction. This is consistent with the explanation above. If the compression noise moves the trajectory to a different minimum rather than degrading it along a fixed axis, then the domain where that minimum is worse can change from run to run.

5.5 In-distribution accuracy does not separate the arms

At 70B no compressed configuration differs from the uncompressed baseline by more than one point, against a difference standard error of 5.0. At 3B the ordering is not interpretable: the uncompressed baseline scores below two of the three compressed configurations, and the range across seeds within a single configuration reaches 5.5 points. Only the INT4-body configuration, which §5.3 shows

⁷The base figure comes from an evaluation-path diagnostic run, not from the adapter sweep, since there is no adapter to name: the un-fine-tuned model was scored on the same 200 WikiText-2 windows of 512 tokens under the same dtype policy, in a process that also reproduced three adapter perplexities to four decimal places.

Table 14: GSM8K accuracy, 8-shot chain-of-thought, greedy decoding with stop sequences and a 512-token budget. 70B evaluates 200 test examples per run and 3B evaluates 200; the binomial standard error is 3.5 points and the standard error of a difference between two configurations is 5.0.

Model	Configuration	n	Accuracy	
Llama-3.1-70B	base model, no adapter	1	91.5	
	uncompressed	1	89.0	
	FP8 head views	1	88.0	
	+ max-abs anchors	1	88.5	
Llama-3.2-3B	uncompressed (C)	2	52.8	± 0.4
	blockwise (A)	8	53.9	± 2.2
	FP8 (B)	3	55.7	± 2.8
	INT4 (D)	3	48.3	± 2.8

corrupts gradients by two orders of magnitude, separates at all, and even there the ranges overlap at 51.0.

We report these numbers because prior work reports them, and because they support the claim that compression is not obviously harmful. They do not support finer distinctions, and §5.7 shows that the distinctions that matter are invisible here.

Our first evaluation used a 256-token generation budget with no stop sequences, under which 98–100% of generations reached the limit. Re-evaluating the same adapters with stop sequences and a 512-token budget reduced truncation to 0–2% at 70B and revealed the actual chain-of-thought length: 57 tokens for the base model and 67–72 after fine-tuning. Accuracy moved by at most 1.5 points, but the re-evaluation *reversed* the ordering of the uncompressed and anchor configurations at 70B (88.0/88.5 $\rightarrow$ 89.0/88.5). We take that reversal as direct evidence that sub-point differences at these sample sizes carry no information, and report the stop-sequence numbers throughout.

At 3B the picture is different in a useful way. Re-evaluating five adapters under the same change altered the number of correct answers by exactly zero, although mean generation length varied from 102 to 364 tokens across them and truncation from 20 to 136 of 200. The generations changed; the extracted answers did not, because the answer string appears within the shorter budget in 153–188 of 200 cases (counted as generations whose answer is extracted from an explicit “The answer is” statement, the `answer_is` source in the evaluation records). Accuracy is insensitive to the evaluation protocol as well as to the damage.

Held-out GSM8K perplexity agrees as well: the four 70B runs we evaluated (two blockwise, one FP8, one uncompressed) are within 0.7%, and at 3B the eight blockwise runs span 0.004 (§5.7). The distinction appears only out of distribution (§5.3).

MMLU was fixed as the primary benchmark before it was run, all 14,042 items at 5-shot, together with two tests: the Spearman correlation between held-out perplexity and MMLU accuracy over the eight blockwise runs, and the mean difference between the four runs above the damage threshold and the four below. Both are null. The correlation is $\rho = +0.31$ at exact two-sided $p = 0.46$ over all 40,320 relabelings, with the sign opposite to the prediction; the damaged runs score 0.32 points *above* the safe ones at $p = 0.26$ over the 70 splits.

The same pre-specification also included a positive control: whether the INT4 arm, which is damaged on every other measurement, separates from the uncompressed pair. We noted there that

without this control, a null inside the blockwise arm would be uninterpretable. That control was never completed. The MMLU chain evaluated the base model, the eight blockwise runs and two of the three INT4 runs, then stopped; the FP8 arm, the uncompressed arm and the third INT4 run have no MMLU records (Appendix E). The two INT4 runs that were evaluated score 55.10% and 56.98% against a blockwise mean of 57.02%, but with no uncompressed cell the comparison has no reference. What shows that the measurement can detect a shift is instead a comparison that was not pre-specified: fine-tuning on GSM8K costs 2.56 points of MMLU on the blockwise arm (the base model scores 59.59% against a mean of 57.02% over the eight runs), and all eight runs fall below the base model (sign test $p = 0.0078$), so the measurement sees a shift of that size without difficulty. We report this as a substitute, not as the control we had planned. Because the FP8 arm was not evaluated on MMLU, this benchmark says nothing about compression’s effect on accuracy between arms; that comparison rests on the four benchmarks of Table 14 and §5.6.

5.6 Aggregate accuracy hides an item-level divergence

Section 5.5 showed that accuracy does not separate the arms. This section looks at the same answer sheets item by item.

Accuracy on these benchmarks is one bit per item, taken from a ranking of four choices, and the ranking is what the arms actually differ on. Because every adapter answers the same items in the same order, two runs can be compared item by item. Two runs from the blockwise arm disagree on 7.39% of the 6,653 items and two FP8 runs on 6.59%, a ratio of 1.12 at exact permutation $p = 0.001$ over all 12,870 relabelings of the sixteen runs. The direction is the same on all four benchmarks, the result is unchanged under length-normalized scoring ($p = 0.0002$), and the worst of the 64 balanced leave-one-out splits, dropping one run from each arm, gives $p = 0.017$.

Two reference points fix the scale. The two uncompressed runs already disagree on 6.48% of items, so most of the churn (the share of items two runs answer differently) is the seed and not the compression; compression adds 0.91 points to that reference, of which 0.80 is what the FP8 rule removes. Fine-tuning itself moves behavior by 13.24 points against the base model, so compression’s whole contribution is 6.9% of what the fine-tuning does.

Table 15: Item-level disagreement between two runs of the same arm, on two item sets: the four benchmarks (6,653 items) and MMLU (14,042), where only the blockwise arm was evaluated. *churn* is the fraction of items answered differently; *gap* is the mean absolute difference in accuracy over the same pairs.

arm	$\cos(g, g^\star)$	pairs	churn (pp)	gap (pp)
<i>four benchmarks, 6,653 items</i>				
uncompressed (C)	—	1	6.478	0.331
FP8 (B)	0.992	28	6.594	0.427
E2M1, blockwise (A)	0.587	28	7.387	0.556
INT4, blockwise (D)	0.371	3	9.079	0.962
<i>MMLU, 14,042 items</i>				
E2M1, blockwise (A)	0.587	28	15.713	0.464

In Table 15 the uncompressed rows are one pair and the INT4 rows are three, so neither is an estimate; the two eight-run arms are the comparison that is tested. The two columns do not move together. Churn grows by a factor of 1.40 from the uncompressed pair to the INT4 arm while the

accuracy gap grows by 2.91, so on these four benchmarks the ratio between them falls monotonically along the gradient-cosine axis: 19.6, 15.4, 13.3, 9.4. A worse configuration does not just disagree more; fewer of its disagreements cancel out. Aggregate accuracy is not blind, only low-resolution: two runs differing on 7.4% of items end up 0.56 points apart, because roughly as many items move one way as the other.

Within the blockwise arm, held-out perplexity still does not say which run is which. Splitting its eight runs at the median perplexity gives a churn difference of 0.01 points at $p = 0.91$. The dispersion is a property of the configuration, and it does not turn into a per-run label on this axis any more than it did on the others.

On MMLU only the blockwise arm was evaluated (§5.5), so the between-arm test cannot be repeated there. Two blockwise runs disagree on 15.71% of MMLU items (28 pairs, 14.65 to 17.05%), more than on the four benchmarks, while their accuracies stay within 0.46 points of each other on average (Table 15). Within the blockwise arm, the split at the median perplexity gives a churn difference of +0.01 points at $p = 1.00$: held-out perplexity does not say which run is which on either item set.

5.7 What the training run does not tell us

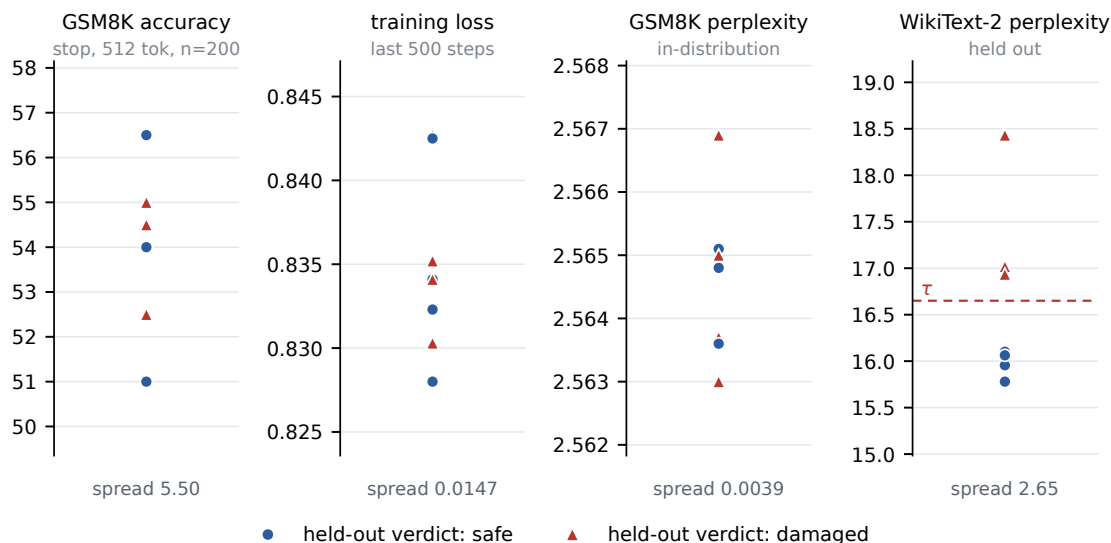


Figure 5: The same eight blockwise runs at 3B under four measurements. Color marks the pre-specified verdict, which is defined on held-out perplexity. Only that panel separates the colors: task accuracy, training loss and in-distribution perplexity interleave the two groups completely. Accuracy is the unified protocol of §4.1; training loss is the mean over the final 500 optimizer steps. The figure under each panel is the spread across the eight runs, in that panel’s units.

The previous section compared arms. This one asks the harder question inside a single arm: given the eight blockwise runs, four of which the threshold calls damaged, does anything observable during training say which four? Every measurement we could make was blind to it (Figure 5), and that matters for how methods of this kind are evaluated, since prior work reports task accuracy and task accuracy is the least sensitive quantity we measured.

Damaged runs average 53.25% against 54.50% for safe ones: the right direction, but not separable (exact two-sided permutation $p = 0.51$). Training loss, averaged over the final 500 optimizer steps, spans 0.828 to 0.843 across the eight blockwise runs; the damaged four average 0.836 against 0.834 for the safe four, a difference of 0.001 with exact two-sided permutation $p = 0.71$. No threshold on training loss separates them, and the ordering by loss does not even go in a consistent direction. The contrast is in scale: loss spans 0.015 across runs whose held-out perplexity spans 2.65.

That comparison still mixes in the data order, since those eight runs differ by seed. The re-run pairs of §5.4 remove even that. Seed 123 ends 2.16 perplexity above its re-run on WikiText-2, and nothing in the two loss traces ever showed it. Over 3,736 steps the windowed RMS of their loss difference stays between 0.005 and 0.010 while the step-to-step variation within either run is 0.13 to 0.18: the separation never reaches a tenth of the noise of the batch the two runs share. It does not build up over time either. Across all three re-run pairs it grows by a factor of only 1.4 to 1.7 between the first hundred steps and the last, so whatever produces the endpoint gap does not show up in the loss as something that accumulates. The blockwise pairs sit about 2.3 times higher than the FP8 pair from the first window onward, but with two pairs against one we do not test this. The claim is about the loss and not about the models: a single number can stay flat while the parameters move apart, and we did not measure the weight-space trajectory.

Pre-clipping gradient norms are strongly format-dependent (Table 16): INT4 bodies average 586 with every step clipped, the blockwise arm averages 1.21 with 75% clipped, and the FP8 arm averages 0.61 with 0.4% clipped, matching the single uncompressed run for which we have a gradient trace (0.59; we added this logging after our other uncompressed runs). Within the blockwise arm, however, the four damaged runs and the four safe runs are indistinguishable:

Table 16: Pre-clipping gradient norm, averaged over training, for the four damaged and four safe runs of the blockwise arm at 3B. The two groups are completely mixed together.

	gradient norm, mean over training			
damaged runs	1.18	1.27	1.25	1.18
safe runs	1.18	1.19	1.24	1.17

The two groups are completely mixed together. Gradient statistics answer “which precision format is this” and not “did this particular run go wrong.”

Evaluated on the held-out GSM8K, the distribution the adapters were trained on, the eight blockwise runs span 2.563 to 2.567, a range of 0.004, and the FP8 runs span a comparable interval. The same eight adapters span 15.78 to 18.43 on WikiText-2. Damage is invisible where the model was trained.

Seed 123 of the blockwise arm shows this directly: it reaches WikiText-2 perplexity 18.43, the highest of the eight and 2.65 above the lowest, while scoring 55.0% on GSM8K, above the uncompressed baseline’s 52.8% rather than below it. Its in-distribution perplexity, 2.5637, ranks fourth of the eight and sits 0.0007 below the arm mean. No non-finite gradient occurred.

The paired and unpaired t agree (§5.4).

As a consequence, in the blockwise arm, half of the runs land at or below the base model before fine-tuning (16.19) on out-of-distribution perplexity and half exceed it by 4.6 to 13.8%. None of the nine measurements we took during or after training, except evaluating on a distribution the model was not trained on, distinguishes the two. What this means in practice is that a damaged run ends

worse on general text than the model it was fine-tuned from (16.19), while every safe run, every FP8 run, and both uncompressed baselines end at or below it. No prior activation-compression method reports run-to-run variation in held-out perplexity in a fine-tuning setting (§2.3).

6 Limitations

Single runs at 70B. One run per length cannot separate a configuration from its seed. The 70B runs also use 500 optimizer steps against 3,736 at 3B, so a 70B accuracy comparison would mix scale with training length. We therefore report 70B as a capability measurement and draw no accuracy conclusion from it. The uncompressed 70B curve at $L=3072$ is extrapolated from $L=1024$ and 1536. The two measured points give a slope of 0.02884 GB per token. The point where that line crosses our budget motivates this paper, but it is not something the paper tests. Two blockwise runs at 70B (seeds 42 and 123) end at WikiText-2 perplexities of 5.94 and 5.90, against 5.93 for the FP8 run and 5.98 for the uncompressed run, and their GSM8K perplexities are within 0.7% of each other. Two runs cannot give a rate, and the 70B runs use 500 steps, a length at which the 3B cohort of Appendix E was unstable in every arm, so we draw no conclusion about the failure at scale from them.

Training length. Every damage rate in this paper is measured after 3,736 optimizer steps at 3B. We also ran the blockwise, FP8 and uncompressed arms at 500 steps, the 70B training length, under a pre-registered rule (Appendix E). All three arms dispersed on held-out text, including one uncompressed run whose WikiText-2 perplexity reached 5,646 while its training loss, gradient norms and in-distribution perplexity stayed normal. The 500-step regime is therefore unstable independently of compression, and the zero-of-eight result for FP8 head views is established at 3,736 steps only. Whether it holds at other training lengths is untested.

The rule over-selects. The rule of §3.2 routes by tensor rank, but the property that actually matters is whether a tensor feeds the softmax bilinear product. Under FlashAttention the two match closely enough to be useful. The rule still promotes V and the rotary tables, which the per-role probe shows do not need it. That costs 0.035 bits per element, or 5.9% of what promoting Q and K costs.

One task, one architecture family. All accuracy results are GSM8K with rank-16 LoRA on the attention projections. Llama-3.2-3B, Llama-3.1-70B and Qwen2.5-3B are all decoder-only transformers with grouped-query attention and RoPE. We have not tested whether the rank-4 rule transfers to multi-query attention, to encoder-decoder models, or to a task with a weaker fine-tuning signal than GSM8K.

The attribution is incomplete. The per-role probe attributes tensors by storage pointer, and re-materialized head views break that chain. This misses 6–16% of head views (7 to 18 of the 56 query views), which therefore stay uncompressed; how many are missed varies between probe runs, and the FP8 point of Table 7 is the least covered. A missed tensor raises the measured cosine. The reported gradient fidelity of the damaged arms is therefore an upper bound, and the gap we attribute to the head views is a lower bound.

The damage count depends on a threshold. We fixed the threshold from the FP8 arm’s first three runs on WikiText-2, after the first eight runs had been evaluated and before the remaining twenty-four (Appendix E). §5.3 reports what happens under the recomputed threshold and under 2σ , 3σ and 4σ definitions. The dose-response tests that use no threshold at all agree with the ones that do. We still report a count that depends on a choice, and we say so wherever the count appears.

No out-of-sample validation. Every dose-response result in this paper is fitted on the same four arms it describes. To make the probe a tool, we would need to predict a configuration nobody has trained, and to register that prediction before training it. We have not run that experiment. We did run its smallest case, and it failed: two arms the probe scores identically ended in measurably different subspaces (§5.1). That result tells us what resolution such a study needs. Two configurations the probe cannot separate still differ by 0.0026 in subspace dispersion. A new arm must therefore exceed roughly three times that to be distinguishable from its neighbor. Given the local slope, this closes the region between cosine 0.587 and 0.992 entirely and leaves two usable positions below 0.587, rather than the four we first planned. Appendix E gives the design in full: what would be predicted, how the arms would be spaced, and the reporting rules. The experiment can then be run as registered rather than reconstructed.

We do not know when the runs separate. The loss traces of two re-runs stay inside batch noise throughout training, while their endpoints differ by 2.16 perplexity (§5.7). We do not know whether the parameters separate gradually or at a few points. Answering that means measuring $\|\Delta\theta(t)\|$ between saved checkpoints. We have those checkpoints but did not analyze them. The two runs of a pair were also checkpointed on different intervals, so the comparison needs the checkpoint steps the two have in common.

The re-run evidence is thin. The claim that damage is not predictable from the seed rests on two deliberate re-runs and one accidental duplicate, and none of the three pairs was run under a frozen code state: every run in Table 23 was launched from a working tree with uncommitted changes (§5.4, Appendix F). The deterministic-mode probe of §5.4 was run from a clean commit and shows that the branch at the third optimizer step disappears when the kernels are made deterministic, but it covers 20 optimizer steps of 3,736. It does not show that a pair which agrees for 20 steps agrees at the end, and we have not trained a full pair under deterministic kernels.

No demonstrated downstream cost. We show that blockwise four-bit compression inflates the variance of out-of-distribution perplexity and of item-level benchmark behavior, and that in-distribution accuracy sees neither. We do not show that a user who releases a high-variance run is harmed. The downstream measurement in §5.5 puts an upper bound on the effect across four multiple-choice benchmarks. It does not demonstrate that the effect exists.

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A Training and evaluation configuration

Table 17 gives every setting that was held fixed. The two scales differ in three places only (checkpoint, training length, and learning rate), and the compression settings are identical, which is what makes the memory results at 3B and 70B comparable.

Across every Llama run in this paper the non-finite counter of §3.3 stays at zero. Qwen2.5-3B is the exception. A four-bit run there at 500 optimizer steps skips 21 steps on non-finite gradients and scores 68.2% (341/500); an earlier four-bit run on the same model scored 0.2% (1/500). We cannot present this as a with-and-without comparison. The earlier run was trained for 3,736 optimizer steps against 500, a factor of 7.5, and it was run before the counter existed, so its non-finite count is unrecorded rather than zero. What the pair establishes is that four-bit training on this model can fail catastrophically and that a run with the safeguard active did not; isolating the safeguard would require a matched control.

The unguarded control of §3.3 (42.4%) is the nearest thing we have to one. Both runs are Qwen2.5-3B, naive_fp4, FP8 rank-4 routing, seed 42, 2,000 examples for one epoch, at the same learning rate and group size. The body encoding is recorded as int4 for the unguarded run and is absent from the guarded one, which ran on 2026-08-17, before the E2M1 encoding and its configuration field existed (commit af8a491 of 2026-08-21 in docs/development_history.txt); INT4 was the only body encoding at that time, so the two agree on it. What they do not share is the code state: the guarded run was launched from commit 849f1a8 and the unguarded one from 1676501, three weeks later, both with uncommitted changes.

The 25.8-point difference is consistent with the guard’s effect and rests on one run against one. The six guarded compressed runs on this model span 68.2% to 75.6%, all above the unguarded figure, and the unguarded run’s training loss over its last 100 optimizer steps is 4.31, against 1.81 for the guarded run and 1.12 to 3.18 for the other guarded compressed runs, which is what applying 21 non-finite steps to the optimizer state would produce. What the pair establishes with certainty is the index relation of §3.3: the guard does not prevent non-finite gradients from arising, only from being

Table 17: Training and evaluation settings. Gradient clipping is recorded in every result file (`grad_clip = 1.0`). Weight decay is not, because it is not configurable in this code path; its value is the constant in `run_experiment.py`. LoRA is attached to the attention projections only, not to the MLP.

	Llama-3.2-3B	Llama-3.1-70B
base checkpoint	meta-llama/Llama-3.2-3B-Instruct	unsloth/...-70B-Instruct-bnb-4bit
base weights	NF4, double quantization, bfloat16 compute	
LoRA rank / α / dropout	16 / 32 / 0.05	
LoRA targets	q_proj, k_proj, v_proj, o_proj	
task	GSM8K	
training examples	7,473, 2 epochs	2,000, 1 epoch
optimizer steps	3,736	500
learning rate	2×10^{-4}	5×10^{-5}
optimizer	PagedAdamW8bit, weight decay 0.01	
schedule	cosine, no warmup	
gradient clipping	1.0, global norm over the LoRA parameters	
batch size / accumulation	1 / 4	
max sequence length	512	
group size g	128	
minimum tensor size	1,024 elements	
data-type policy	§3.3: bfloat16 norms, embeddings, head and adapters	
accuracy, in training	500 problems (A/B seeds 789–4042: 200)	200 problems
accuracy, unified re-eval	200 problems, 8-shot, 512 new tokens, stop sequences	
perplexity	200 WikiText-2 chunks, 500 GSM8K problems, 512-token cap	

Table 18: The unguarded and guarded Qwen2.5-3B runs of §3.3.

	guard	GSM8K	non-finite steps
qwen_safeguard_control	off	212/500 (42.4%)	21
qwen3b_skip_check	on	341/500 (68.2%)	21

applied. Isolating its effect on accuracy would need matched runs at several seeds under one code state, which we have not done.

Two further Qwen2.5-3B records need a caveat. First, two runs of the configuration that stores every tensor at FP8 (`uniform_fp8`), made on 2026-08-17 and differing in recorded configuration only by output directory, scored 40.0% and 70.8% and flagged 20 and zero non-finite steps. An earlier draft of this paper read the pair as run-to-run variation on a second model. The record does not support that reading: the 40.0% run holds 500 step losses against 20 flagged steps, whereas every run in which the guard skipped steps holds one loss fewer per skipped step (479 for 21 skips), so its 20 flagged steps were not skipped that way. What was done with them is not recoverable from the record; the run’s last-100-step training loss of 5.26 and its score both resemble the unguarded control (4.31, 42.4%) rather than any guarded run. The pair shows that whether overflow occurs at all varies between runs of one configuration on this model, and nothing more. Second, the guarded compressed runs end with a last-100-step training loss of 1.12 to 3.18 while the uncompressed run at the same settings ends at 0.37, and their GSM8K scores are alike (68.2–75.6% against 68.6%). We

have no explanation for the gap in loss and report it as unexplained.

Of the three models of §4.1, Qwen2.5-3B-Instruct has activation outliers an order of magnitude larger than either Llama model and triggers a failure mode the others do not (above). For 70B we load the publicly available pre-quantized checkpoint `unsloth/Meta-Llama-3.1-70B-Instruct-bnb-4bit`. Online NF4 quantization of the 132 GB BF16 shards accumulates host memory faster than it releases it and does not complete within our budget; the pre-quantized checkpoint also holds the quantization step constant, which helps reproducibility.

The container cap and the abort ceiling of §4.1 exist because without them memory pressure makes the host unresponsive. The effective budget is therefore somewhat below the nominal 128 GB, and we state measured peaks rather than extrapolating from them wherever a measurement was possible.

Training uses no padding and the PagedAdamW8bit optimizer (Dettmers et al., 2023); Table 17 lists the remaining settings.

Sequence lengths at these training settings are short in practice: GSM8K examples tokenize to 168 tokens on average (maximum 468), well below the 512 cap. Activation memory is correspondingly a small fraction of the total, and packing overhead can exceed the saving at such lengths: at 70B the compressed and uncompressed curves cross at $L \approx 486$ (§4.3). We therefore separate the two questions. Memory and throughput are measured on synthetic fixed-length sequences, where the length is controlled exactly; accuracy is measured on natural-length GSM8K. Reporting memory at natural lengths would understate the method, and reporting accuracy at synthetic lengths would measure nothing.

All arms of Table 2 share the same interception mechanism and filter order and differ only in how precision is allocated. The uncompressed arm retains its activations in bfloat16; where it is reported alongside encoding variants, the same measurement is reused.

The stop rule of §4.1 terminates generation at the next question boundary.

The number of seeds per arm at 3B varies with what each claim requires and is stated in every table; the two arms the central claim rests on use eight, {42, 123, 456, 789, 1011, 2024, 3033, 4042}. At 70B, compute constraints limit us to a single seed, which we note where relevant.

Tensors are partitioned along the trailing dimension into groups of $g = 128$ channels. Each group is scaled by its own maximum absolute value, rounded to a signed four-bit integer and bit-packed two codes to a byte; the scale is kept in FP16, adding $16/g = 0.125$ bits per element for a total of 4.125. A group whose length is not a multiple of g is padded to the boundary, and the padding is stripped on dequantization so that no padded element reaches a gradient. The unpack path reverses this in one pass: unpack nibbles, sign-extend, multiply by the group scale, and reshape to the stored shape.

Table 19 removes the two float32 activation sources of §3.3 one at a time, in a single session so the three cells share a process.

Tables 20 and 21 are the two tables behind §4.3. Both compare compression against gradient checkpointing on the same hardware and the same batch shape; neither is a claim about which is preferable, only about what each costs.

At $L=2048$, the longest sequence at which the uncompressed baseline runs at all, Standard LoRA exceeded our 100 GB ceiling while four-bit compression used 76.85. Extrapolating both curves exhausts the 128 GB budget at $L \approx 3000$ without compression and $L \approx 5800$ with it, both roughly twice the lengths we measured. The two protected encodings differ in memory but not in time: 62.77

Table 19: Removing the two float32 activation sources, one at a time, in a single session. Llama-3.2-3B, $B=2$, $L=4096$, NF4 base weights, 100 steps after 10 warm-up. Each correction saves two bytes per stored element, as the tensor shapes predict: 2.82 GB over 56 normalization sites against 2.82 predicted, and 5.72 GB over 112 adapter sites against 5.64. Neither costs time; step time falls by 5%.

RMSNorm	LoRA adapters	Peak (GB)	Step time (s)
float32	float32	51.91	13.53
bfloat16	float32	49.09	13.18
bfloat16	bfloat16	43.37	12.81

Table 20: Memory and step time at Llama-3.2-3B, $B=2$, $L=4096$, NF4 base weights, 100 steps after 10 warm-up, AdamW. Measurement runs use the default AdamW; our training runs use 8-bit paged AdamW. The optimizer state differs by 0.055 GB for a 9.2M-parameter adapter, under 0.13% of any cell.

	Peak alloc. (GB)	vs. base	Step time (s)	vs. base
uncompressed	43.37	1.00×	12.99	1.00×
recomputation	18.54	0.43×	17.02	1.31×
four bits, Q/K at FP8	30.15	0.70×	15.29	1.18×

against 61.06 GB reserved at $L=1024$ in the timing session of Table 21 (INT4 body; the E2M1-body session of Table 3 gives 63.23 against 61.64), at 42.76 against 42.29 seconds on one run each. Each series in Figure 2 comes from one session; peak reserved varies by 0.2 to 1.7 GB between sessions on the compressed arms and by nothing measurable on the uncompressed one.

B Arm definitions

An arm is a triple: training method, body encoding, and rank-4 mode. All three are needed to identify it. An uncompressed run records a body encoding and a rank-4 mode as well, neither of which affects it, so resolving an arm from those two fields alone silently merges the uncompressed arm into the FP8 arm (B) wherever the two share an output directory, and they do share one. Resolving by directory is therefore also wrong, since no directory holds exactly one arm. Table 22 gives the triple for each arm. Run-level records, including the output directory of every run, are in the released repository (see the code link on the first page).

Table 21: Step time at Llama-3.1-70B, $B=1$, one session per length. The $L=1024$ column averages 15 steps after 3 warm-up; $L=3072$ averages 7 after 3, which is adequate for a memory reading but marginal for timing, so we quote its ratio as consistent with the $L=1024$ figure rather than as an independent measurement. Standard LoRA does not fit at $L=3072$.

	$L=1024$	vs. base	$L=3072$
uncompressed	40.12	1.00×	—
recomputation	57.20	1.43×	166.10
four bits, Q/K at FP8	42.76	1.07×	125.89
four bits, Q/K per-channel	42.29	1.05×	—

Table 22: The six arms as configuration triples. *body* is the encoding applied to tensors of rank two and three; *rank-4* is the encoding applied to the four-dimensional head views. The triple rather than the output directory is what counts.

arm		method	body	rank-4	n
A	blockwise	naive_fp4	E2M1	blockwise 4-bit	8
B	FP8	naive_fp4	E2M1	FP8	8
C	uncompressed	standard	—	—	2
D	INT4	naive_fp4	INT4	blockwise 4-bit	3
E	INT4-body variant	naive_fp4	INT4	FP8	3
F	per-channel	naive_fp4	E2M1	per-channel INT4	8

C The per-role probe

Fix a checkpoint and a batch. Let $g^\star$ be the gradient computed with no compression and $g_{\mathcal{F}}$ the gradient computed with compression applied only to the tensors in role $\mathcal{F}$. We report the cosine similarity between the two over the LoRA parameters. Each measurement is a single forward-backward pass, so a sweep over roles costs minutes rather than the hours a training run would take. We did not log the probe’s wall-clock time. The two probe sessions behind Tables 4 and 7 were started seven minutes apart (19:16:55 and 19:24:07 by the machine’s clock on 2026-09-04), the first of them running five filters, and that gap is the only timing record we have; the five minutes of the title is that order of magnitude, not a measurement.

Attributing tensors to roles is not simple: the query, key, and value projections read the same post-normalization activation, and rotary embedding and repeat_kv re-materialize their inputs, breaking pointer identity. We tag storage pointers at the projection outputs and again after each re-materializing operation, which reaches 84–94% of the 112 head-view tensors a forward pass retains at 3B (56 query views, 28 key views and 28 value views). The key and value views are tagged completely in every probe. The query views are not: across the probe files behind Figure 3 and Tables 4 and 7, the Q/K filters reach 38 to 49 of the 56, the count varies between otherwise identical probe runs, and Table 7 gives it for each of its four points; the remainder carry a pointer that a re-materialization re-labels, and pass through uncompressed. The gap works against our claim, not for it: the cosine of 0.36 we report below is measured with 7 of the 56 query views left intact, so compressing all of them would lower it further. Between arms it is not neutral: the FP8 point is the least covered, 40 of 56, so of the four dose–response cosines it is the one most biased toward one.

The attribution is incomplete, and the error goes in one direction only. A head view whose tag is lost is not compressed, so it contributes an uncorrupted gradient and raises the measured cosine. The 0.36 reported for Q and K is therefore an upper bound on their fidelity under compression, and the gap against the 0.99 of the other roles is a lower bound.

The shares in Figure 3 are computed over the role-attributed elements that the probe observes, counted the way autograd stores them. The six roles sum to 80.6%. The remaining 19.4% is the normalized residual, which each of the three attention projections saves once ($3 \times 6.3\%$), plus the up_proj input and the o_proj output. The RMSNorm input is likewise saved twice per layer. The training runs do not deduplicate saved tensors, so each of these copies is packed and stored separately and the shares are shares of what is stored. Logits and rotary tables carry no role attribution and lie outside the denominator.

D Every run

Table 23 lists all 35 full-length Llama-3.2-3B runs: the 32 that enter the arm aggregates and the three re-trainings discussed in §5.4. It is generated from the result files by `scripts/make_appendix_tables.py` rather than transcribed, and the eight per-arm value sets it produces were checked against the values plotted in Figure 4; all six arms agree. The adapter path of each run is in the released repository’s run-level records (see the code link on the first page). The three re-trainings marked with a dagger are excluded from the arm aggregates: they are second trainings of a seed that was already trained, discussed in §5.4, and counting them twice would understate exactly the variance that section is about. Arms are resolved from (method, body encoding, rank-4 mode) jointly; the one row marked ‡ was run before those fields were recorded, and its arm is inferred from the output directory recorded in the released run-level records.

E Design for an out-of-sample validation

This design is registered but not run. We record it here so that the study can be executed as specified rather than designed after the fact. The constraint that sets its size came from a prediction of our own that turned out to be wrong.

We call a quantity *pre-specified* when it was written into the repository before the data it is applied to had been evaluated, and *registered* only when that record also carries a timestamp we do not control. Three quantities in this paper are pre-specified in this sense, two of them only in part, and none of them is registered.⁸ The damage threshold of §5.3 was fixed after the first eight runs (three of the blockwise arm, three of the FP8 arm, both uncompressed) had been evaluated and before any of the other 24 were, so it is pre-specified for 24 of the 32 runs it classifies. The prediction for the INT4-body variant of §5.1 was committed before its target was computed, but only 33 seconds before. The MMLU primary metric, primary comparison (Spearman over the blockwise arm) and secondary comparison (damaged against safe) of §5.5 were committed before any MMLU cell ran; the clause declaring the four cheaper benchmarks confirmatory was committed 21 minutes after the chain had started running them, and the copy of the script the chain took at launch does not contain it. The same chain also scheduled HellaSwag as a second stage; its cells were killed by a memory-guard misconfiguration after partial work and were not re-run, so the paper reports the four benchmarks that completed. The 22-adapter MMLU chain (`results/mc_mmlu_launch.log`) evaluated the base model, the eight blockwise runs and two INT4 runs and was not resumed, so the FP8 arm, the uncompressed arm and the third INT4 run have no MMLU records; every MMLU statistic in the paper is therefore over the blockwise arm, the base model, or the two INT4 cells. Every other MMLU statistic was defined on the four benchmarks first and repeated on MMLU, and is marked that way where it appears. The program below is the only thing we call registered, and it counts as registered only if it is pushed to a public host before any of its runs is trained.

⁸All orderings rest on the local commit history and file times; no external registration service was used. The commit hashes below belong to the private development history, which the released repository does not carry; its commit log is archived as `docs/development_history.txt` in the released repository. The threshold header of `scripts/run_axis2_a4dfp4_seeds.sh` is commit 75b756b (2026-08-31 20:01 UTC); it lists the first three blockwise perplexities, two of them above the threshold. The INT4-body prediction is commit 2479f1a (2026-09-08 09:41 UTC), 33 seconds before its target was first computed. The MMLU header of `scripts/run_mc_chain.sh` is commit 27ae5b6 (2026-09-07 17:05 UTC), 14 hours 37 minutes before the first MMLU cell ran.

Table 23: Every full-length Llama-3.2-3B run behind the paper’s aggregate statistics: 7,473 examples, two epochs, 3,736 optimizer steps. WikiText-2 and GSM8K perplexity are token-level over 200 chunks and 500 problems at a 512-token cap. Accuracy is the unified re-evaluation at 200 problems with stop sequences, recomputed as $100 \times n_{\text{correct}}/n$; a dash means the adapter was not in that re-evaluation. A dagger marks the three re-trainings excluded from the arm aggregates; a double dagger marks the one run that was run before the configuration fields were recorded, whose arm is inferred from its run-level records. Arm letters are those of Table 2.

	arm	seed	WikiText-2	GSM8K ppl	acc. (%)
	A	42	15.7796	2.5636	56.5
	A	789	15.9558	2.5636	51.0
	A	2024	16.0627	2.5651	54.0
	A	1011	16.1016	2.5648	56.5
	A	4042	16.9350	2.5650	54.5
	A	456	16.9550	2.5630	51.0
	A	3033	17.0183	2.5669	52.5
	A	123	18.4320	2.5637	55.0
†	A	123	16.2705	2.5681	—
†	A	4042	16.3508	2.5678	—
	B	1011	15.8887	2.5591	—
	B	42	15.9331	2.5570	57.0
	B	123	16.0008	2.5582	57.5
	B	4042	16.0253	2.5602	—
	B	3033	16.0370	2.5580	—
	B	789	16.0927	2.5570	—
	B	456	16.2828	2.5557	52.5
	B	2024	16.3029	2.5560	—
†	B	789	15.9989	2.5591	—
	C	42	16.0717	2.5586	52.5
	C	456	16.0812	2.5573	53.0
	D	123	17.2634	2.6535	48.5
‡	D	42	17.6943	2.6401	45.5
	D	456	33.7466	2.6309	51.0
	E	42	15.9250	2.5585	—
	E	456	16.0782	2.5585	—
	E	123	16.4676	2.5607	—
	F	3033	15.9908	2.5591	—
	F	456	16.1257	2.5614	—
	F	4042	16.2414	2.5642	—
	F	123	16.2876	2.5609	—
	F	42	16.3664	2.5578	—
	F	1011	16.5298	2.5590	—
	F	2024	16.5877	2.5567	—
	F	789	16.6579	2.5604	—

The claim to be tested is ordinal, not quantitative: where a new arm falls in the ranking, not what its dispersion equals. The four existing arms do not support a quantitative fit. Leaving each arm out in turn and predicting it from a least-squares line through the other three misses the INT4 arm (D) by 6.8 to 19 times its internal range and the blockwise arm (A) by 3.3 to 10 times, under every one of thirty coordinate systems (Table 24). The per-channel and FP8 arms (F and B) land anywhere from less than a tenth to forty times their range, depending on the transformation. The transformation that does best overall still misses the INT4 arm by 6.8 times. The thirty systems are six transformations of the cosine axis crossed with five of the dispersion axis, with the prediction mapped back before the error is taken; the full list is in `scripts/prereg_loo.py`. A quantitative prediction would not hold up. The planning document states this range as three to ten times with a best case of 3.2; that figure was computed without a committed script and does not reproduce, and the table replaces it.

Table 24: Leave-one-arm-out prediction of subspace dispersion from gradient cosine, with error in units of the held-out arm’s internal range (max minus min over its within-arm pairs). Each held-out arm is predicted by a least-squares line through the other three, fitted separately in each of thirty coordinate systems. A transformation diverges when its error is non-finite or above 50; min, median and max are over the remaining transformations.

held-out arm	cosine	min	median	max	diverging
D (INT4)	0.371	6.8	14.2	19.3	0
A (blockwise)	0.587	3.3	8.1	10.3	0
F (per-channel)	0.910	0.08	3.8	11.2	0
B (FP8)	0.992	0.5	6.6	40.2	7

The primary target is subspace dispersion, computed per module and averaged by arm exactly as in §5.4: its within-arm range is 0.002 against between-arm gaps of 0.011 to 0.029, and it stays tight at three runs. The secondary is item-level disagreement, which matters because it is behavioral and rules out the objection that the prediction targets a quantity nobody uses. Log standard deviation of held-out perplexity is registered third and registered as weak: at four seeds its relative standard error is 41%, so neighboring arms may swap order. Damage count is not registered at all, since it depends on the threshold.

The resolution of the measurement, not our preference, fixes how many arms can be added. Requiring three times the 0.0026 residual gives $\Delta \cos \geq 0.057$ below cosine 0.6, $\Delta \cos \geq 0.23$ between 0.6 and 0.9 where the slope flattens, and $\Delta \cos \geq 0.058$ above 0.9. Against the existing arms at 0.371, 0.587, 0.910 and 0.992, the two upper windows are too narrow and two positions remain, both below 0.587. The two must come from different interventions rather than one knob turned twice; candidates are protecting Q or K alone, a layer threshold, group size, and the head-view grid. Group-size arms move bits per element as $16/g$ and so may not enter the bits-held-fixed contrast of §5.3.

Every arm trained is reported whatever it shows, at four seeds fixed at registration, and a failed run is repeated and reported. A secondary target that succeeds where the primary fails is not relabeled as primary, and a region the probe cannot reach is reported empty.

One further pre-registered experiment was run and is reported here because its rule closed it. The header of `scripts/run_axis5_short_training.sh` fixed, before the data arrived, a test of whether the blockwise damage is a function of training length: the blockwise arm at eight seeds

and the FP8 arm at four seeds were trained for 500 optimizer steps (2,000 examples, one epoch, the 70B training length), and the variance ratio of their WikiText-2 perplexities was to be read against $F > 8.89$ (damage present at 500 steps) and $F < 5$ (the arms disperse alike). The outcome was $F = 2.19$, and it is uninterpretable rather than negative: the FP8 arm, which is the stable reference at 3,736 steps (sd 0.151), is itself dispersed at 500 steps (sd 6.10; runs at 15.38, 15.73, 18.03 and 28.36), the blockwise arm runs from 15.54 to 42.09, removing one extreme run from each arm brings F to 1.06, and the median-centered Brown–Forsythe statistic is $W = 0.0055$ against a critical value of 4.96. A second header (`scripts/run_axis5_cstd_then_70b_seed123.sh`) then fixed a rule for the uncompressed arm at 500 steps and four seeds: if any seed exceeded a GovReport perplexity of 100, or the WikiText-2 spread matched the four-bit arms, the 500-step regime would be judged unstable independently of compression and the route closed. Two of the four seeds did: seed 123 reached 116.3 on GovReport, and seed 456 reached 5,646 on WikiText-2, 2,655 on NarrativeQA and 26,097 on GovReport, while its training loss (0.99 at step 500), gradient norms (at most 0.90, no non-finite step) and in-distribution GSM8K perplexity (2.624, within 0.004 of the other three seeds) gave no sign of it. The route was closed as the rule required. The per-run records are in `results/ppl_axis5` and `results/axis5_short`. What this experiment establishes is narrow: at 500 steps, held-out perplexity is unstable with or without compression, and in-distribution perplexity does not see it in either case; the damage rates of §5.3 are therefore statements about 3,736 steps.

F Reproducing the reported statistics

Table 25: Every MMLU statistic in the paper, and when each test was fixed. *pre-specified*: in the chain header at commit 27ae5b6, before any MMLU cell ran. *four-benchmark*: defined on the four cheaper benchmarks in §5.5 and applied to MMLU unchanged. *exploratory*: chosen after seeing MMLU. The positive control was pre-specified as a comparison without a statistic and could not be completed, because the uncompressed arm was never evaluated on MMLU; the fine-tuning cost that stands in for it was not pre-specified. A is the blockwise arm (Table 2). All values are recomputed by `scripts/audit/mmlu_stats.py` from the MMLU records in the released repository.

test	value	p	status
Spearman(ppl, acc), arm A, 8!	$\rho = +0.31$	0.46	pre-specified
damaged – safe, arm A, 70 splits	+0.32 pp	0.26	pre-specified
INT4 – uncompressed	not available	—	pre-specified (comparison only)†
all 8 arm-A runs below base, sign test	8/8	0.0078	not pre-specified‡
churn, damaged/safe split, 70 splits	+0.01 pp	1.00	four-benchmark
Mantel churn $\times \Delta\text{ppl} $, 8!	$\rho = +0.14$	0.67	tab:levels
Mantel $ \Delta\text{acc} \times \Delta\text{ppl} $, 8!	$\rho = -0.17$	0.35	tab:levels
within-damaged – within-safe churn	+1.38 pp	0.029	exploratory

† The header fixed the comparison (INT4 against uncompressed) and no statistic; the uncompressed arm has no MMLU records, so it was never run.

‡ The substitute control, used because the pre-specified one could not be completed.

Four rules determine which numbers enter which statistic. They are stated here because each one of them, applied wrongly, produces a plausible answer rather than an error.

Arm membership is determined by the training method rather than by the directory a measurement landed in: as §B sets out, membership is (method, body encoding, rank-4 mode) jointly, since an

uncompressed run records a body encoding and a rank-4 mode too, and neither affects it. Dropping the method puts the uncompressed run at seed 42 into the FP8 arm as a ninth run, which changes the WikiText-2 variance ratio from 33.66 to 38.47.

Accuracies recorded during training are not comparable across seeds, because the protocol changed between cohorts: some runs were scored on 500 GSM8K problems and others on 200, with different stop-sequence handling. Every arm was therefore re-evaluated afterwards under one protocol, and all reported accuracies come from that directory. Within it, accuracy is recomputed as $100 \times n_{\text{correct}}/n$ rather than read from the stored percentage.

Three adapters are second trainings of a seed that had already been trained: the blockwise arm at seeds 123 and 4042, and the FP8-arm duplicate at seed 789. They are separate runs, not repeated measurements of one run, and §5.4 is about precisely that. They are excluded from the arm aggregates, and marked with a dagger in Table 23, because including both members of a pair would shrink exactly the variance that section reports. Where a pair must be reduced to one value (Table 11 uses the FP8 arm at seed 789, a configuration that was trained twice by accident), the later run (16.093) is taken. Every run in Table 23 was launched from a working tree with uncommitted changes: `env.git_dirty` is true in each of its records, as it is in 143 of the 146 run records that carry the field and were made before the determinism probe of §5.4, so the commit hash in such a record locates its code state only approximately. The 16 records of that probe are the exception: each was launched from a tree that was exactly one commit, and the flag is false in all of them. In the same table, the NarrativeQA and GovReport values for seeds 42, 123 and 456 of both arms come from a separate evaluation pass, stored under `results/pp1_ood_extra` in the released repository.

The damage threshold is $\tau = 16.65$, fixed as the uncompressed mean, 16.08, plus three standard deviations of the FP8 arm as it was at the time, 0.19 over its first three seeds. It is not the value you get from the final data: the eight FP8 runs have standard deviation 0.1510, which gives 16.5293. Every count in the paper is at 16.65. §5.3 reports where that matters, which is the per-channel arm alone: at the recomputed value it flags three runs instead of one, and one of the three exceeds it by 0.0004. At 2σ , 3σ and 4σ the threshold is 16.46, 16.65 and 16.84; the blockwise arm flags the same four runs at all three, because its empty band runs from 16.10 to 16.94.

Under those rules the reported statistics are recomputed as follows, from Table 23 and nothing else (Table 26).

Two choices behind the tests of §5.3 are explained here rather than there.

Welch’s t is designed for unequal variances, and the variances of the blockwise and FP8 arms differ by a factor of 33. The F -test assumes normal samples. With 59% of the blockwise arm’s variance carried by one run, it puts the variance ratio at $p = 1.4 \times 10^{-4}$ two-sided, a factor of 211 below the permutation value of 0.0303 (390 of 12,870 relabelings, also two-sided). The correction applies to the variance tests and not to the means, and swapping the two procedures in the other direction does not work either. A permutation test asks whether the two samples are exchangeable, so with variances differing by a factor of 33 it can reject on the dispersion alone and can no longer be read as a statement about means. It is therefore listed above as a robustness check, $p = 0.065$ for the blockwise-arm mean. For the per-channel arm, whose variance ratio is 2.37, the permutation p for the mean matches Welch’s 0.015 to three decimals.

Under a single normal distribution, the largest-gap statistic gives $p = 0.07$ for the arm, and the largest gap is in any case the one isolating the single worst run at 18.43 rather than the one separating

Table 26: Each statistic in §5.3 and §5.4 recomputed from the per-run values in Table 23. Arm letters are those of Table 2. The variance ratio is the ratio of sample variances with the larger on top; its p is two-tailed on $F_{7,7}$. Brown–Forsythe is Levene’s test centered on the median.

statistic	definition	value
arm A mean, sd	8 runs	16.6550, 0.8759
arm B mean, sd	8 runs	16.0704, 0.1510
arm C mean, sd	2 runs	16.0764, 0.0067
mean difference	A – B	+0.5846
Welch t	unequal variances	1.86
p	two-tailed Welch t , quoted	0.103
p	exact permutation, robustness only	0.0650
variance ratio	s_A^2/s_B^2	33.66
p	two-tailed $F_{7,7}$, not quoted	1.4×10^{-4}
p	exact permutation on $ \log F $	0.0303
Brown–Forsythe W	median-centered	9.64
p	parametric, not quoted	0.0078
p	exact permutation	0.0185
τ	16.08 + 3(0.19), pre-specified	16.65
damaged at τ	D, A, F, B, C	3/3, 4/8, 1/8, 0/8, 0/2

the groups. A test that fixes the split at four-four gives $p = 0.03$, but it fixes the split at a position chosen by looking at the data. With eight points, no test reaches significance unless it fixes the split position in advance, so §5.3 reports the separation as an observation.

The determinism probe of §5.4 is the one experiment in this paper launched from a clean commit. One configuration is trained twice for 20 optimizer steps on the paper’s data order (the same 7,473 examples, two epochs and permutation; only the step count is limited), once under PyTorch’s default kernels and once under `torch.use_deterministic_algorithms` with a fixed cuBLAS workspace and the flash attention backend, each launch in its own process. Every record stores each micro-step loss, a sha256 of every LoRA gradient before clipping and a sha256 of every LoRA parameter after the update, at every step, and the losses of steps 1 and 2 reproduce Table 10 in every launch. Table 27 gives the first step at which the two launches of a pair differ. Under the default kernels the gradients differ from the first backward pass while the forward losses of that step agree, the parameters differ from the second update, and the loss from the third step; under deterministic kernels nothing differs. Of the kernels of the training step, run thirty times each on one input, only the flash-attention backward returns different bits under the default kernels (one bf16 ulp), and it repeats under deterministic mode; the matrix products, the LoRA chain, cross-entropy, RMSNorm, the gate activation, dropout, the NF4 dequantization, the 8-bit optimizer step, gradient clipping and our pack and unpack repeat under both. The op-level record, the run records, the chain with its prediction written before the first launch, and the script that produces the table are listed in the manifest.

Table 27: The determinism probe. Two launches of one configuration at the seed shown, 20 optimizer steps, under PyTorch’s default kernels and under deterministic mode. Each entry is the first optimizer step at which the two launches differ in the sha256 of the LoRA gradients (before clipping), in the sha256 of the LoRA parameters (after the update) and in the loss; “none” is bit-for-bit agreement through step 20. Step time is the mean over steps 2 to 20 of both launches, in seconds. Produced by `scripts/audit/determinism_compare.py` -latex from `results/determinism/`.

configuration	kernels	gradients	parameters	loss	s/step
blockwise four-bit (A), seed 123	default	1	2	3	2.34
	deterministic	none	none	none	2.75
blockwise four-bit (A), seed 4042	default	1	2	3	2.35
	deterministic	none	none	none	2.76
FP8 head views (B), seed 123	default	1	2	3	2.29
	deterministic	none	none	none	2.69
uncompressed (C), seed 123	default	1	2	3	1.90
	deterministic	none	none	none	2.15